

FANUC America Corporation SYSTEM R-30iA, R-30iB, and R-30iB Plus Controller RemovalTool Setup and Operations Manual

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Version 7.20 and Later

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FANUC America conducts courses on its systems and products on a regularly scheduled basis at its headquarters in Rochester Hills, Michigan. For additional information contact

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Patents

One or more of the following U.S. patents might be related to the FANUC America products described in this manual.

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VersaBell, ServoBell and SpeedDock Patents Pending.

Conventions

This manual includes information essential to the safety of personnel, equipment, software, and data. This information is indicated by headings and boxes in the text.

**Warning**

Information appearing under **WARNING** concerns the protection of personnel. It is boxed and in bold type to set it apart from other text.

**Caution**

Information appearing under **CAUTION** concerns the protection of equipment, software, and data. It is boxed to set it apart from other text.

Note Information appearing next to **NOTE** concerns related information or useful hints.

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Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system-including the design, implementation, and debugging of application programs-must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and **not** replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.

- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points* . Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and *work envelope* to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.



Warning

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:



Warning

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.
- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that an applicator bell cup can continue to spin at a very high speed even if the robot is idle. Use protective gloves or disable bearing air and turbine air before servicing these items.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.



Warning

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.

- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

**Warning**

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

Note Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

OVERVIEW

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RemovalTool allows you to perform surface and edge finishing applications with a wide variety of FANUC robots using FANUC-supplied force control devices. RemovalTool software is an option and is used with the HandlingTool software.

RemovalTool supports the following kinds of force control devices:

- Active wrist
- Passive wrist
- Passive bench

RemovalTool software contains macros that can be used to communicate with the active force control devices. For example,

- Commands to set and modify applied forces
- Automatically weigh the tool

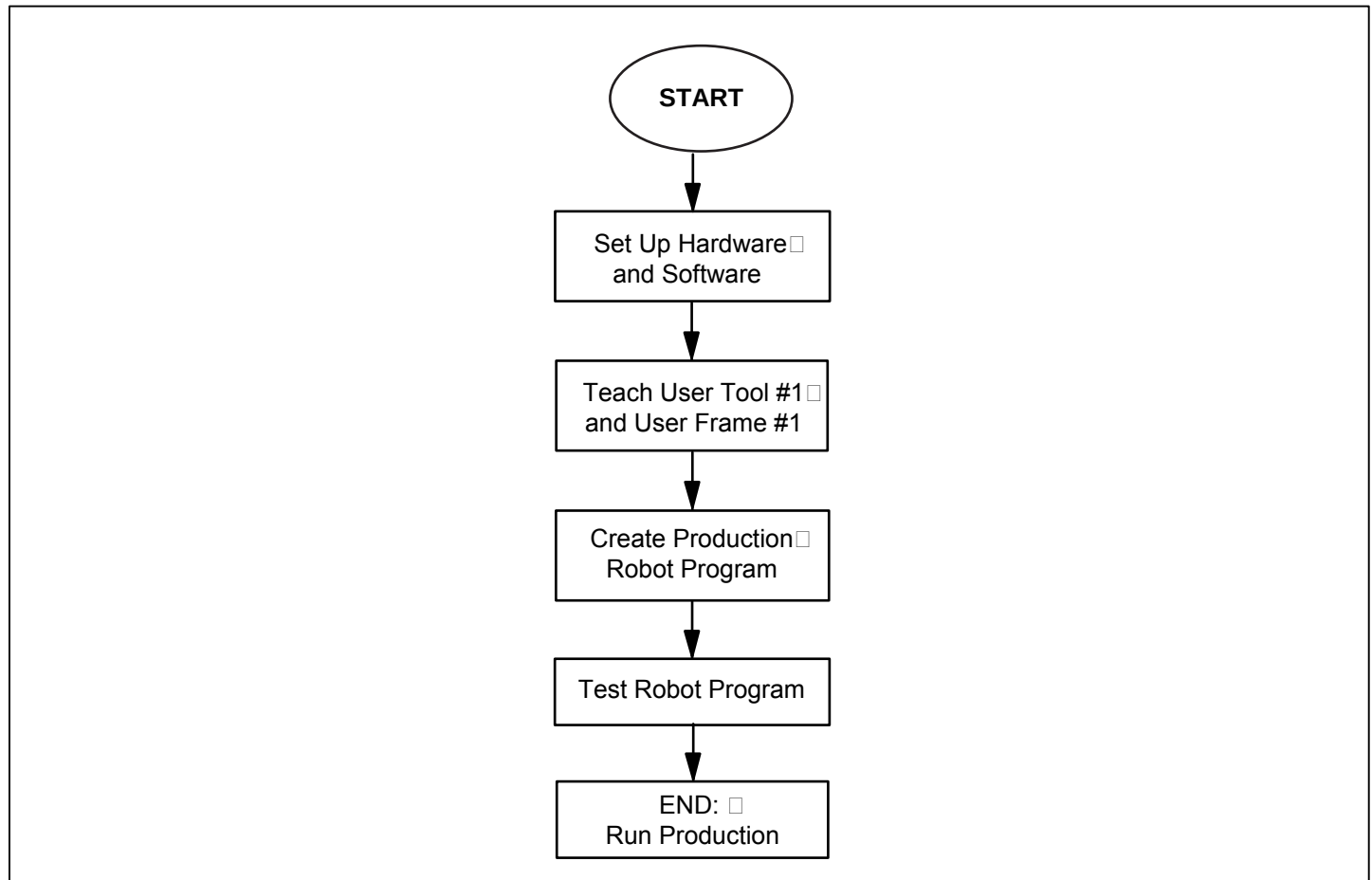
It also contains two other useful functions for passive force devices:

- **Wheel wear compensation** to compensate for the wear in the abrasive wheel during productions.
- **Tool weight compensation** for passive devices to compensate for forces changing due to changing wrist orientation of the robot. It also compensates for gravity.

Teach pendant template programs are also provided for you to get started quickly and to understand the order in which the RemovalTool macros are used.

1.1 SYSTEM OVERVIEW

This section contains a flowchart describing the sequence of operations you must follow in order to set up and use a RemovalTool application. See [Figure 1-1](#) .

Figure 1–1. RemovalTool System Overview

1.1.1 Setup

RemovalTool software allows you to work directly with external devices such as force control devices and pneumatic spindles, using macros. The equipment must be set up before you begin your application. Other kinds of setup include tools and frames. Certain numeric and position registers must also be set up prior to operation.

Refer to [Chapter 2 SETTING UP REMOVALTOOL](#) for information on setting up RemovalTool.

1.1.2 Programming

An application program is a combination of instructions that, when executed in a sequence, will complete your edge or surface finishing application. Predefined macro instructions communicate with force devices and accomplish common tasks encountered in a force control system.

Refer to the "General Setup" and "Program Elements" chapter in the *FANUC America Corporation SYSTEM R-J3iC Controller HandlingTool Setup and Operations Manual* for more information.

Template Programs

Refer to the program example that corresponds to the force control device you are using. The following three teach pendant programs are included with the RemovalTool software:

- Active wrist program *ACTVWTMP.TP*
- Passive wrist program *PASSWTMP.TP*
- Passive Bench program *PASSBTMP.TP*

Refer to [Appendix A](#) for more information about these template programs.

1.1.3 Testing

After you have set up RemovalTool and, if necessary, modified the template programs provided by FANUC America Corporation, you must test your application to be sure it runs correctly. Refer to [Section 4.1](#) for more information about testing your RemovalTool application.

Testing the application is an important step in creating a successful application. Be sure to test your program thoroughly before running it in production.

1.1.4 Running Production

Running production is the final step in executing your application. After you have set up RemovalTool and tested your program, press the CYCLE START button on the controller and the robot performs the RemovalTool operation. Refer to [Section 4.4](#) for more information about running your RemovalTool application in production.

1.2 TERMS

Become familiar with the terms in this section in order to use RemovalTool effectively.

Force Control Devices

You must specify the kind of *force control device* used for the removal process. RemovalTool supports the following kinds of devices:

- Active wrist
- Passive wrist

- Passive bench

Other devices require approval by FANUC America Corporation before they can be used with RemovalTool.

User Frame

Defining the *user frame* is required for the bench mount force control devices. It is used for wheel wear compensation calculations. Use the **three point method** to define the user frame.

Tool Frame

Defining the *tool frame* is required for wrist mount devices. Tool Frame #1 must be used. It is used for wheel wear and tool weight gravity compensation calculations. FANUC America Corporation recommends that you use the **six point method** to determine both tool offset and tool orientation when you define the tool frame.

Command Force

The *command force* is the force applied by the tool medium to the contact point on the surface of the workpiece. The RemovalTool software automatically takes into account the tool weight and tool orientation, and maintains the command force.

Tool Medium

The *tool medium* is the abrasive medium used during removal. Tool media can be of wheel, disc, or cup type. RemovalTool needs to know the type of medium used to determine if it can execute and calculate wheel wear. RemovalTool does not calculate wheel wear for disc type media.

Tool Weight

The *tool weight* is the weight of the current tool in use. It is used to compensate for tool weight against gravity for any tool orientation. The tool weight can be determined manually or automatically, depending on the force control device being used. You must provide accurate weight information for the tool device being used. This includes the weight of the following:

- Motor
- Tool medium
- Slide on which the motor is mounted
- Any other device used to mount the tool to the force control device

The active wrist force control device contains an option that weighs the tool automatically. This should be used whenever possible. For passive devices, the tool weight must be determined manually.

Force/Toolweight Compensation

Force compensation applies to only passive wrist mounted devices. The command force in an application is specified for the vertical orientation of the tool. In order to maintain the specified force between contact points, the force value must be modified at tool orientations other than the vertical orientation.

Tool Wear Compensation

Tool wear compensation applies to all the devices: active wrist, passive wrist, and passive bench mount. As the tool (abrasive wheel diameter for example) wears out, it needs to be detected and the Tool Frame must be modified by the tool wear amount. RemovalTool software provides a macro for calculating wheel wear.

SETTING UP REMOVALTOOL

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2.1 HARDWARE INSTALLATION

You must install the force control devices before you can use them. Installation consists of the following:

- Mounting a motor
- Mounting the force device
- Pneumatic connections
- Active force device controller

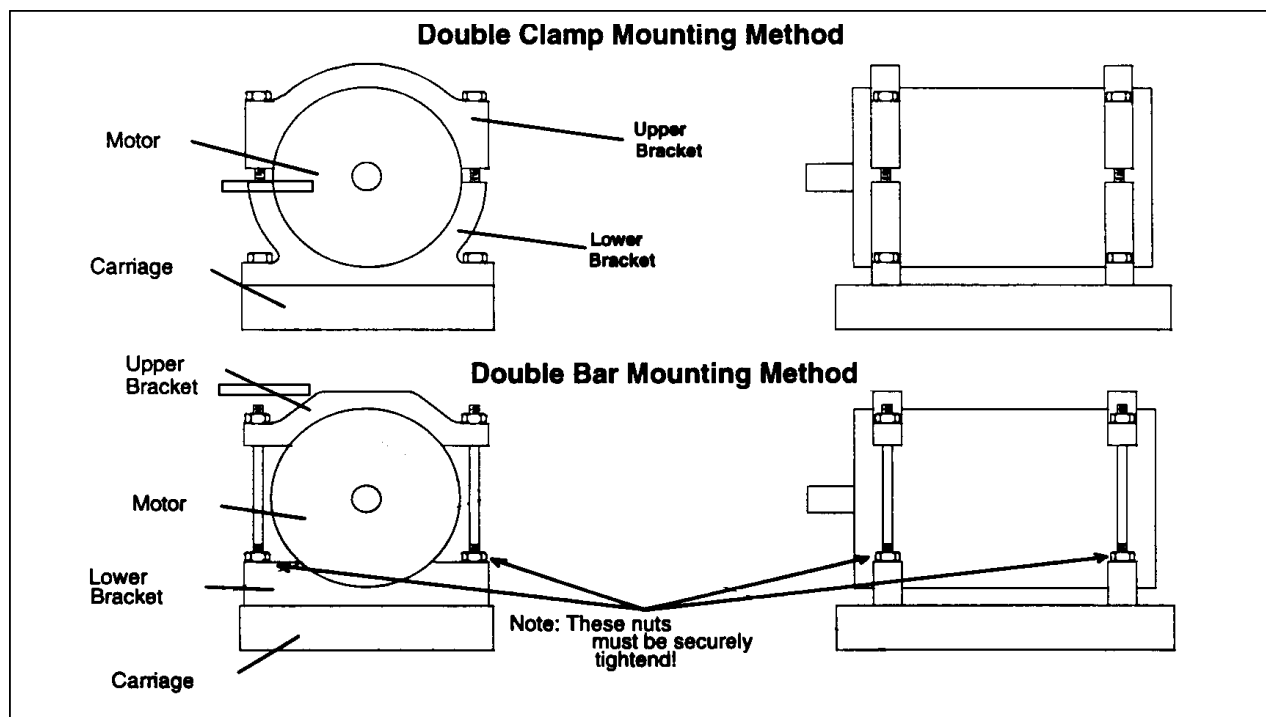
2.1.1 Mounting the Motor

RemovalTool is compatible with various types of motors.

The motor shaft must be oriented parallel or perpendicular to the carriage axis.

When you mount a motor to the carriage, and you are using bracket other than those provided by FANUC America Corporation, make sure that the design of the brackets will enhance process capability by increasing the stiffness of the carriage. Incorrectly designed brackets can deform the carriage mounting plate, causing problems with the internal slide rail alignment.

The force control devices require a double clamp method of mounting. See [Figure 2-1](#) . Even though this method is difficult to fabricate, it is nearly impossible to achieve less than optimal results when using it. The cap screws secure the bottom clamp to the carriage. This clamp acts as a stiffening rib. Tightening the upper cap screws forces the upper clamp against the motor, providing additional stiffness to the carriage plate.

Figure 2–1. Double Clamp Mounting Method

The bolt pattern on the carriage plate is designed to facilitate motor installation in a number of configurations, providing secure attachment points that will not vibrate, causing the bolts to loosen.

**Caution**

Do not use cap screws that extend beyond the threaded depth or damage WILL occur.

2.1.2 Mounting the Force Device

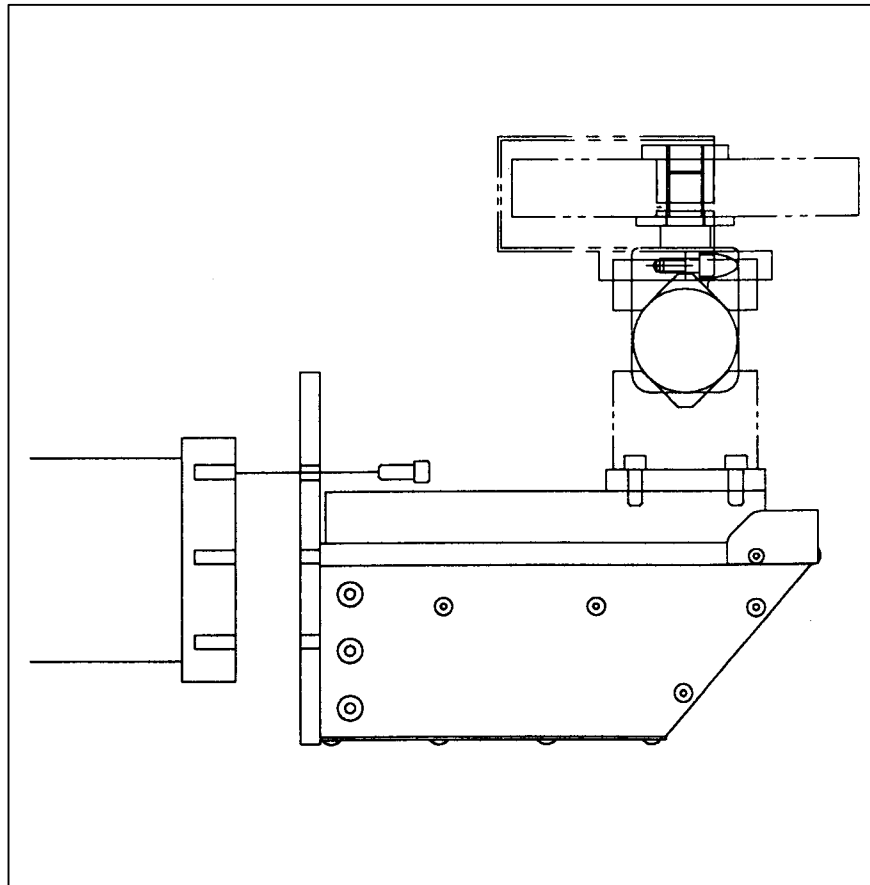
The force control device will be mounted differently, depending on which type of device you are using. The force control device can be directly attached to a robot, or fastened to a bench or floor stand. However you choose to install it, mount the device and its tool so that the positive “+” direction

of the device is pointing to the extended direction and the negative “-” direction corresponds to the retracted direction.

Active Force Control Device

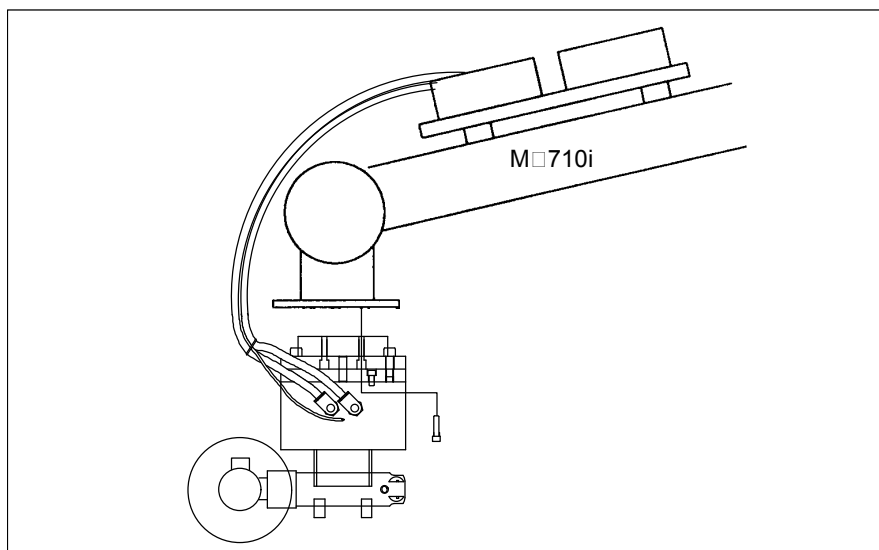
The active force control device uses a three hole mounting pattern. A minimum of three cap screws are required to attach the device securely to a fixture or robot tool mounting plate. See [Figure 2-2](#)

Figure 2-2. Active Force Control Device Installation



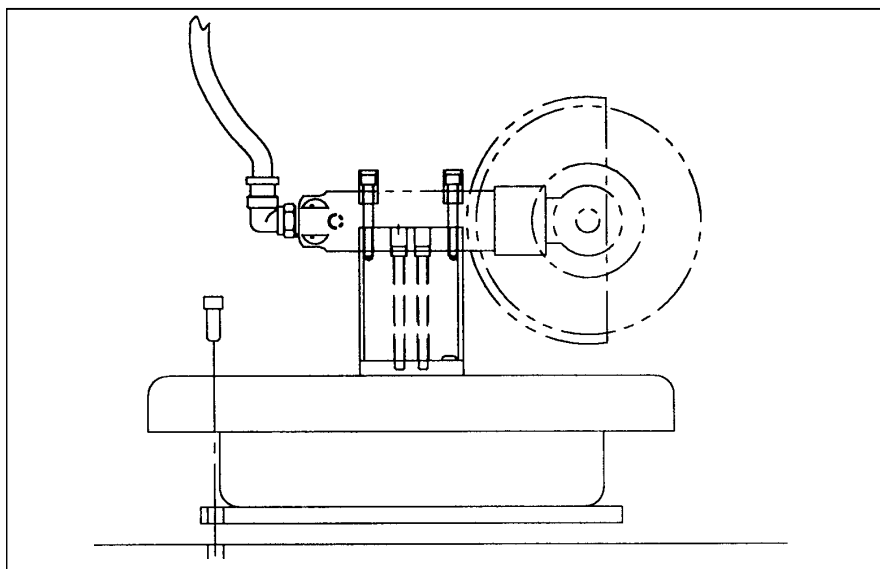
Passive Force Control Device

Mount the passive force control device, via the mounting plate, to the robot. See [Figure 2-3](#) . The control boxes are designed to mount to a typical FANUC robot arm. The pneumatic lines should not interfere with robot movement.

Figure 2–3. Passive Force Control Device Installation**Passive Bench Mounted Device**

The passive bench mounted device should be fastened to a bench or floor stand using the four .27 diameter mounting holes, located one on each corner of the base plate.

The mounting plate for the passive bench mounted device should be centered on the top surface of the device. See [Figure 2–4](#)

Figure 2–4. Passive Bench Mounted Device Installation

Note When you mount the tool, make sure that electrical or pneumatic supply lines do not restrict the applied force motion. If possible, suspend the supply lines above the unit to minimize restricting forces.

2.1.3 Pneumatic Connections

In order to run properly, both the active and passive force control devices require the filtering of clean, dry air. For optimal system performance the air pressure should be maintained at 90 psi, with a minimum and maximum pressure tolerance of 80 psi and 100 psi respectively.

If water condensation is a problem in your air supply system, an air dryer device is required. Moisture inside the device will cause pneumatic failure.

The force control devices maintain a positive air pressure within the device housing to impede the infiltration of contaminated materials. It is important to provide a continuous compressed air supply to the device if the work environment is laden with air born contaminants. If the environment is clean during non-operational periods, you can shut off the air supply to the device using the standard shutoff valve.

Actuators Special Handling

The actuators used in the force control devices are designed and constructed to assure low friction and lead free operation.



Warning

Do not attempt to manipulate the pistons or cylinders. Otherwise, you could injure personnel or damage equipment. If you are having a problem with these parts, contact FANUC America Corporation or the original equipment manufacturer.

The unit does not require lubrication. Lubrication will cause the unit to fail.



Warning

Do not pressurize the actuator prior to installation in the equipment. Otherwise, you could injure personnel or damage equipment.

Active Force Control Device

The active force control device requires a dry, 5 micron filter and unlubricated air supply with an optimal pressure of 90 psi. Filtered air is recommended in high speed flow control devices, used internally to the device, since they have very small orifices and cannot tolerate any foreign material in

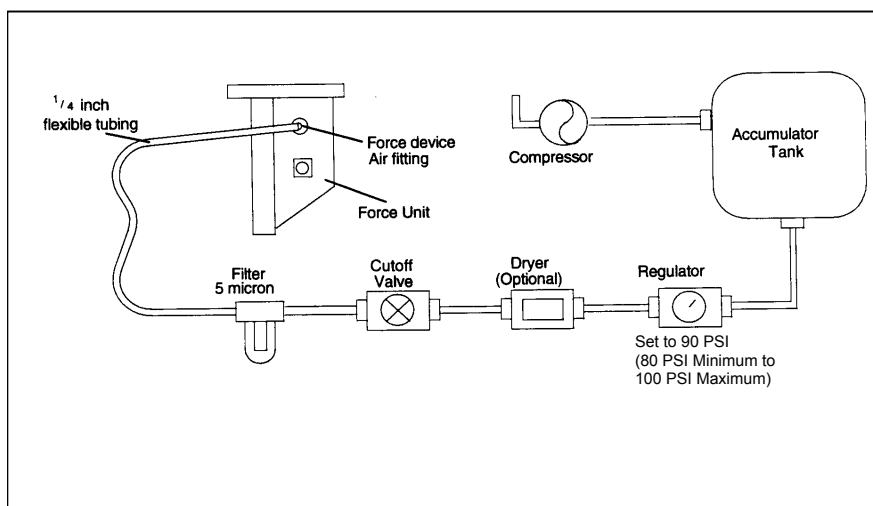
the air supply. The 5 micron filter is required to remove any particles in the air supply system. This filter should be located just before the line going to the force control device.

A minimum air pressure of 80 psi and a maximum air pressure of 100 psi must be maintained for the device to operate properly. Excessively low or high air pressures will cause inferior force control performance and instability.

The single supply line to the device is 1/4 inch in diameter and made of flexible nylon tubing. You should route the tubing to the device, such that there are no kinks and plenty of slack to allow for robot motion. Before you insert the air supply tubing into the force control device air fitting, open the supply valve to blow out any contaminants that might be in the line. Then push the supply line tubing into the self locking fitting. Charge the system with compressed air. Verify that there are no air leaks and that there is an air supply of 90 psi (Pressure tolerance: 80 psi minimum, 100 psi maximum).

The pneumatic supply system should be configured as shown in [Figure 2-5](#)

Figure 2-5. Active Force Control Device Pneumatic Connections



Passive Force Control Device

The required air supply must be filtered clean, dry air (filtered to 5 microns or better), regulated at 90 psi. A minimum air pressure of 80 psi and a maximum air pressure of 100 psi must be maintained for the device to operate properly.

Make sure that the pneumatic supply lines do not restrict the applied force motion. Suspend the supply lines above the unit to minimize restricting forces.

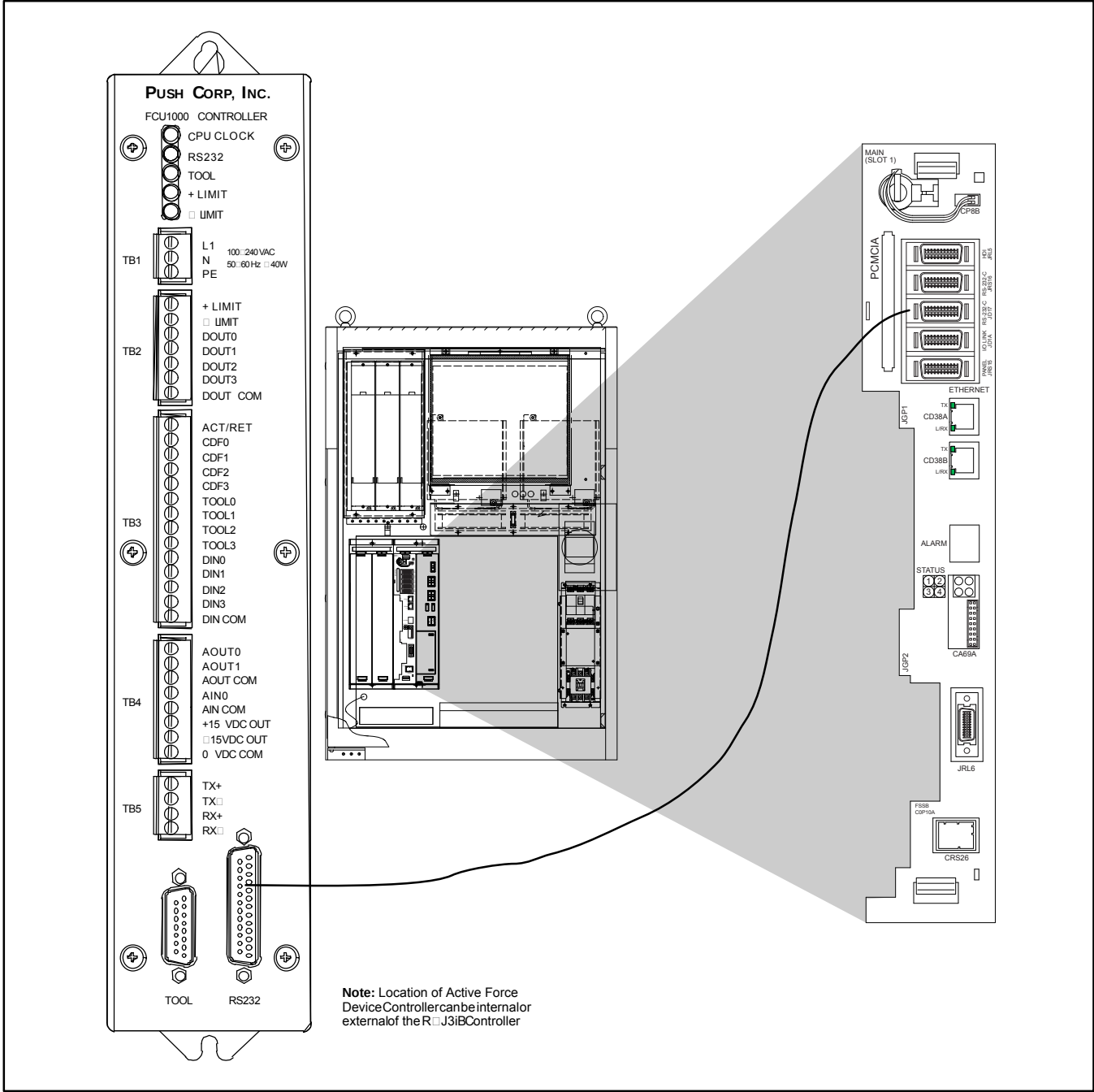
Passive force devices, such as AFD50 and AFD70 Series devices, do not come with built-in electro-pneumatic valves. It is recommended that you use external electro-pneumatic regulators such as SMC Valve (Part No. ITV2050-11N3N4) with 0-20 mA current range. One ITV regulator is needed for each air port used on the force device. To operate these regulators, you can purchase an analog output card (ADA02A, FANUC Part No. A05B-2300-J460) for FANUC model A I/O rack.

Refer to the Pushcorp Device Manual for information on how to use a Positive Pressure Air Supply 20-30 psi for purging.

2.1.4 Active Force Device Controller

The active force device controller is mounted inside the controller and requires one connection to serial port JD17 located on the CPU controller. See [Figure 2-6](#)

Figure 2–6. Active Force Device Controller



2.2 PORT SETUP FOR ACTIVE FORCE DEVICE CONTROLLER

Before you set up RemovalTool for the first time, you must verify that the P4 port setting is set to No Use. This is the serial port on the controller that is connected to the force controller. If this setting is wrong, RemovalTool software will not be able to communicate with the force control device. Use [Procedure 2-1](#) to set up a port.

Procedure 2-1 Setting Up a Port

Conditions

- The default device is set.

Steps

- Press MENU.
- Select SETUP.
- Press F1, [TYPE].
- Select Port Init. You will see a screen similar to the following.

```

SETUP Port Init
      Connector      Port      Comment
1 JRS16 RS-232-C    P2:    [Maintenance Cons]
2 JD17  RS-232-C    P3:    [KCL/CRT          ]
3 JD17  RS-422      P4:    [                      ]

```

- Move the cursor to the port you want to set up and press F3, DETAIL. You will see a screen similar to the following.

```

SETUP Port Init
JRS16 RS-232-C      P2:
  1 Device          [Maintenance Cons]
  2 Speed(Baud rate)      [9600 ]
  3 Parity bit        [None ]
  4 Stop bit          [1bit  ]
  5 Time out value(sec)    [  0]

```

- Select each item and press F4, [CHOICE], to select the appropriate value.

Note To indicate that you are not using a port, set the port to *No Use* . Set the port to no use if you are performing a READ/WRITE from a KAREL program.

7. A device cannot be assigned to two ports. To move a device to another port, set the existing port to **No Use** and then assign the device to another port.
8. Perform a Cold start of the controller to implement the changes to the Port Init screen.

2.3 PERCH POSITION

RemovalTool provides a program called PRGPERCH.TP which is intended to move the robot to a home position in the cell. The home position must be taught in PR[10] for PRGPERCH.TP to work. It is expected that PRGPERCH.TP will be called before executing the Adjust UTOOL macro for wheel wear calculation.

2.4 NOMINAL UTOOL

If your application uses wrist-mounted devices, you must teach a nominal UTOOL frame for each individual tool mounted to the force control device. It is highly recommended that you use the 6 point method of teaching the UTOOL frame.

Refer to the "General Setup" chapter of the application-specific *HandlingTool Setup and Operations Manual* for more information on setting up tool frames.

Gravity Compensation

The gravity compensation feature can only be used with passive devices that are able to apply force in opposing directions while using two cylinders. This feature uses the register assignment R[6] to set the direction of the slide. To operate properly, the positive slide direction of the passive wrist-mounted force device must be along the UTOOL frame -x direction when R[6]=0, or along the +z direction when R[6]=1. See [Figure 2-7](#) and [Figure 2-8](#) for detailed illustrations.

If your application has two tools mounted to your device, where one tool operates in the extended position and one tool operates in the retracted position, you need to change the UTOOL frame and direction by modifying the register assignment R[7] when changing from one tool to another.

Note

R[7]=0: Reverse force is off.
R[7]=1: Reverse force is on.

Wheel Wear Compensation

The wheel wear calculation feature can be used with an active or a passive device, and it uses the register assignment R[15] to set the direction of the slide. It is also necessary to copy the UTOOL value into position register PR[1: Nominal Utool].

Note For the wheel wear calculation feature to work on a passive device, a linear potentiometer must be installed inside the unit to monitor the carriage position.

For the wheel wear calculation feature to operate properly, the positive slide direction must be along the UTOOL frame +x direction when R[15]=1, along the +y direction when R[15]=2, or along the +z direction when R[15]=3. See [Figure 2-7](#) and [Figure 2-8](#) for detailed illustrations.

Figure 2-7. X Axis Slide Orientation

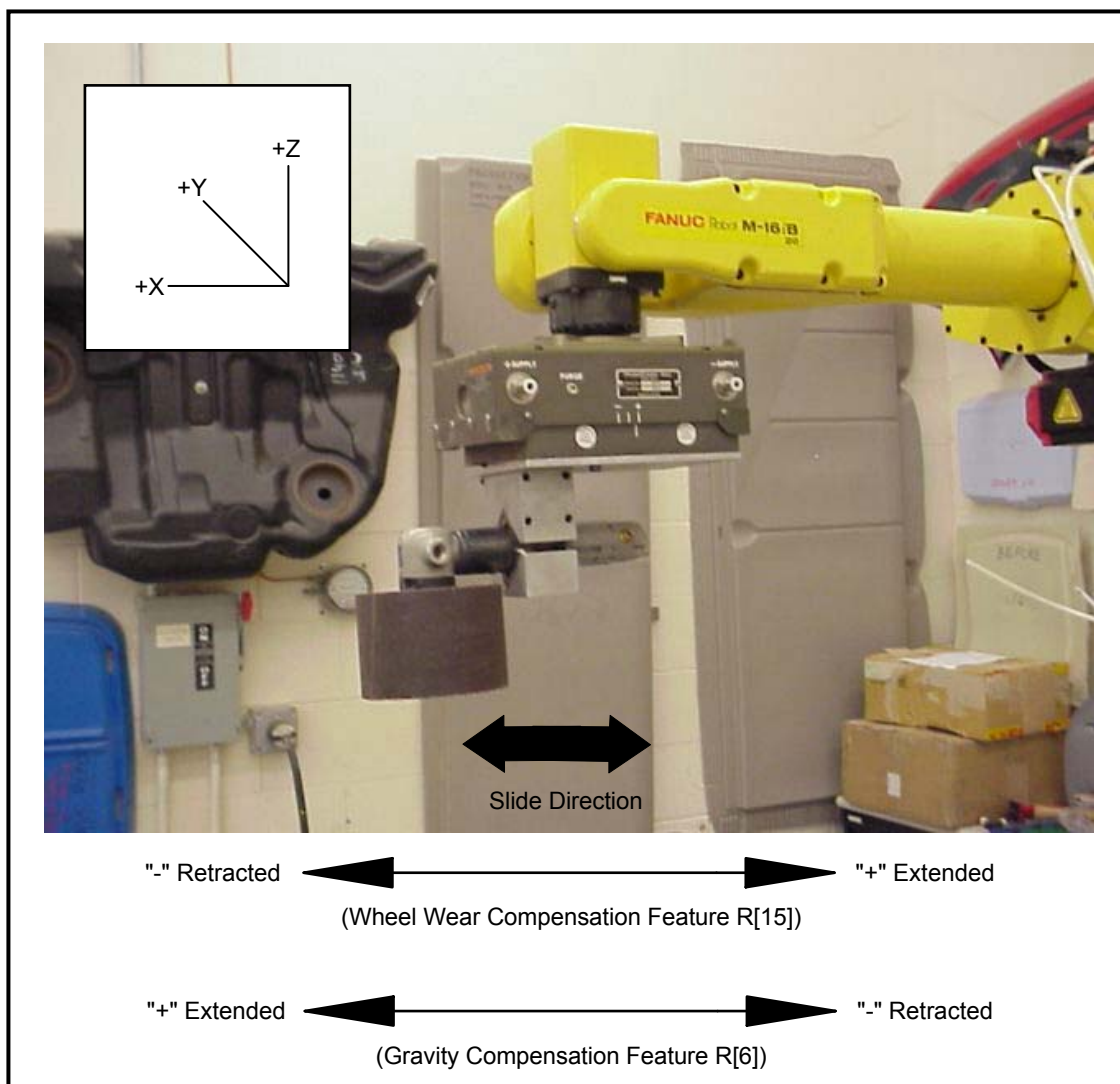
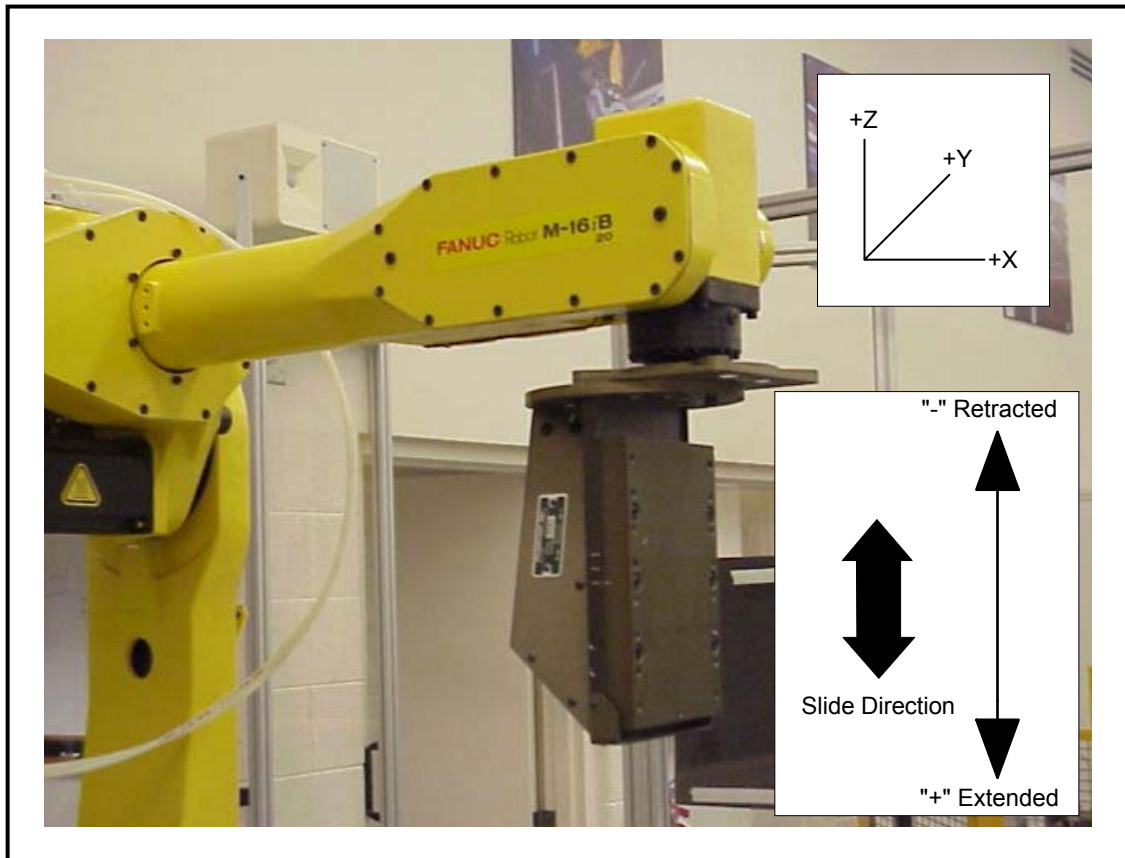


Figure 2–8. Z Axis Slide Orientation



2.5 NOMINAL UFRAME

If your application uses bench mount devices, you must teach a nominal UFRAME which is required by the Adjust UFRAME macro used for wheel wear calculation.

After you have done this, copy the UFRAME value to PR[7:Nominal Uframe].

Refer to the "General Setup" chapter of the application-specific *HandlingTool Setup and Operations Manual* for more information on setting up user frames.

2.6 SEARCH DISTANCE AND DIRECTION FOR WHEEL WEAR

The Adjust UTOOL and Adjust UFRAME macros use PR[2:Incr Search dist] to move the robot incrementally towards the part. The search direction is specified by entering values for x, y, z in PR[2]. For example, if you want to have a search direction parallel to the z in the robot world coordinates,

you would enter -2 mm for z and zeros for x, y, w, p, r. This would make the robot move downwards towards the part in increments of 2 mm when the Adjust UTOOL macro is executed.

2.6.1 Search Start Position

In order to determine wheel wear, teach a search start position using the guidelines below.

Then, periodically during production, call the program containing the search start position and the Adjust UTOOL or Adjust UFRAME macro.

Figure 2–9 illustrates how to teach the search start position using the active wrist. When you teach the search start position, make sure that the slide direction is perpendicular to the surface of the part.

Figure 2–9. Teach Search Start Position - Active Wrist

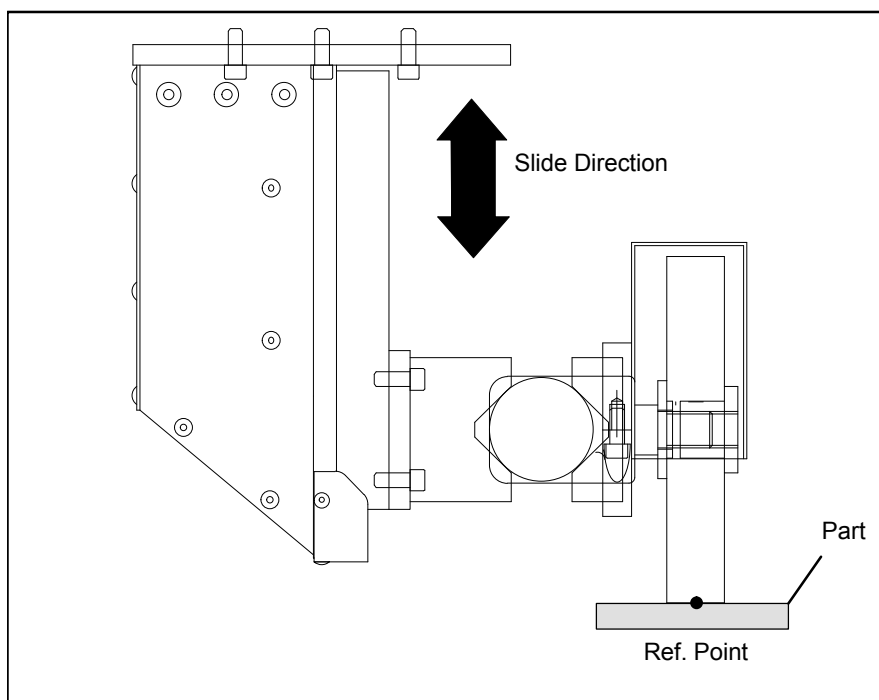
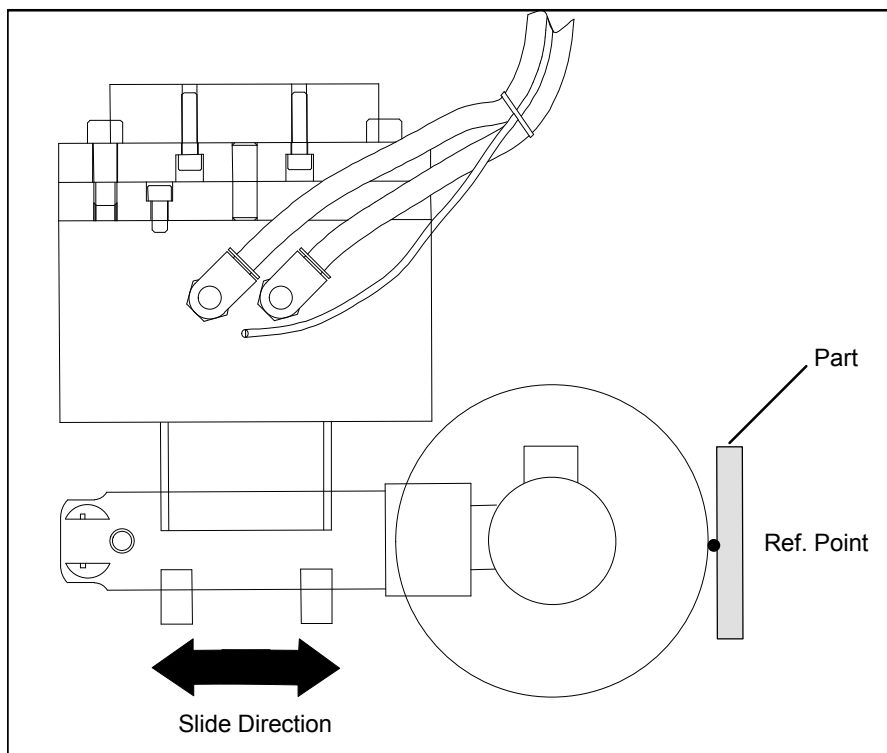
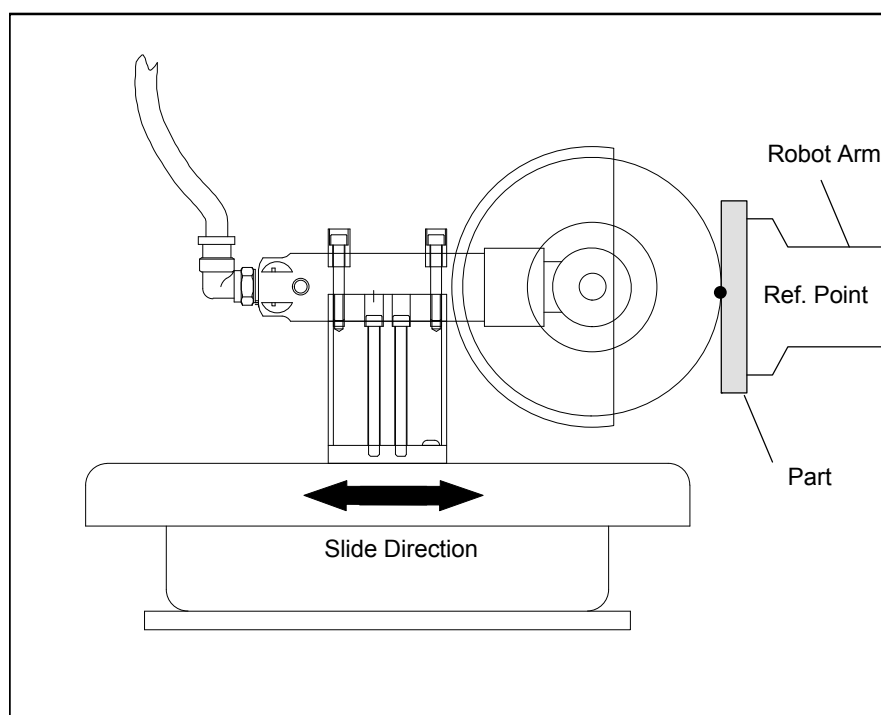


Figure 2–10 illustrates how to teach the search start position using the passive wrist. When you teach the search start position, make sure that the slide direction is perpendicular to the surface of the part.

Figure 2–10. Teach Search Start Position - Passive Wrist

[Figure 2–11](#) illustrates how to teach the search start position using the bench mounted device. When you teach the search start position, make sure that the part is perpendicular to the wheel or disc, and at the center point of the wheel or disc.

Figure 2–11. Teach Search Start Position - Passive Bench Mount Device

2.7 FORCE DEVICE

During RemovalTool setup you must indicate what kind of device you are using. R[4:Force Device] is provided for this purpose. Type a value into R[4] based on the information in [Table 2–1](#) :

Table 2–1. Force Device Setup

Force Device	R[4] Value
Active Wrist	1
Passive Wrist	2
Passive Bench	3

Refer to the application-specific *HandlingTool Setup and Operations Manual* for more information on setting registers.

2.7.1 Passive Force Device Weight Compensation Program

You can adjust the MR_WCOMP.PC macro to different kinds of passive devices: AFD2000, AFD72 and AFD52. MR_WCOMP.PC (pc_equate) contains a force compensation routine that has variables you set for the particular model of the passive device.

The structure for the force compensation routine is shown in [Figure 2–12](#).

Figure 2–12. MR_EQUATE Structure for Force Compensation Routine

```

TYPE
  mr_equate = STRUCTURE
    use_new_equ  : BOOLEAN
    bias_voltage : REAL
    fp           : REAL
    vp           : INTEGER
    fn           : REAL
    vn           : INTEGER
    fnet         : REAL
    fp_pram1     : REAL
    fp_pram2     : REAL
    fn_pram1     : REAL
    fn_pram2     : REAL
  ENDSTRUCTURE

use_eq IN CMOS : pc_equate

```

Tool Weight Compensation program (MR_WCOMP.PC) uses its subroutine (MR_EQUATE.PC) for force compensation. MR_EQUATE.PC can be modified to fit specific models of force devices. This makes RemovalTool software flexible for passive devices.

You can modify FP_PRAM1, FP_PRAM2, FN_PRAM1, and FN_PRAM2 to accomplish this, where the letter “p” stands for positive force variables, and the letter “n” stands for negative force variables.

Refer to [Procedure 2-2](#) to adjust the parameters to adjust MR_EQUATE.PC to each type of device

Procedure 2-2 Adjusting Parameters in the MR_EQUATE.PC Macro

1. Press SELECT.
2. Select the program MR_EQUATE and press ENTER.
3. Press DATA and press F1, [TYPE].
4. Select KAREL Vars.

Note You must have the KCL option loaded in order to access the KAREL variables.

At initialization, the array shown in [Figure 2-12](#) is set to zero, except for the following variables. The default values are shown in [Table 2-2](#).

5. Change the fp_pram and pn_pram for each type of passive device accordingly. Refer to [Table 2-2](#).

Table 2-2. Default Values for Passive Device Parameters

Passive Device	fp_pram1	fp_pram2	fn_pram1	fn_pram2
AFD2000 *	20.52	21.19	18.43	20.17
AFD72	18.27	0	18.27	0
AFD52	2.61	0	2.61	0
AFD-82	36.54	0	36.54	0

These settings are based on Push Corp's force device manuals and can only be used with the SMC ITV2050-11N3N4 external electro-pneumatic regulator.

* The AFD2000 does not require an external electro-pneumatic regulator.

2.8 TOOL WEIGHT

RemovalTool software needs the tool weight information to compensate for gravity and orientation changes from the nominal position.

If you have an active wrist mount device, you can use the Auto Weigh macro to determine the weight of the end of arm tooling (EOAT). This value is stored in R[10] and is communicated to AFD by the set tool weight macro. Set Tool WT macro is included at the beginning of any program as shown in [Figure A-1](#).

If you have a passive wrist mount device, then you must manually determine the tool weight in pounds (lb.) and enter the weight into R[10]. Two decimal accuracy is sufficient. For example, R[10]=26.41 lb. You must include the following items in your tool weight calculation:

- Weight of the motor and clamps
- Weight of the tool medium
- Weight of the device carriage slide

Table 2–3.

Model of Device	Carriage Weight = Wc (lbs)*
AFD-52/51	0.3
AFD-72/71	3.5
AFD-82/81	9

Note *

1. The carriage weight is taken from the PushCorp manuals of corresponding devices.
2. Subject to change at OEM's discretion at any time.
3. The carriage weight needs to be added to the tooling weight which will be attached to the carriage.

After you have determined the weight, it must be entered into R[10:Tool Weight]. Refer to the application-specific *HandlingTool Setup and Operations Manual* for more information on setting registers.

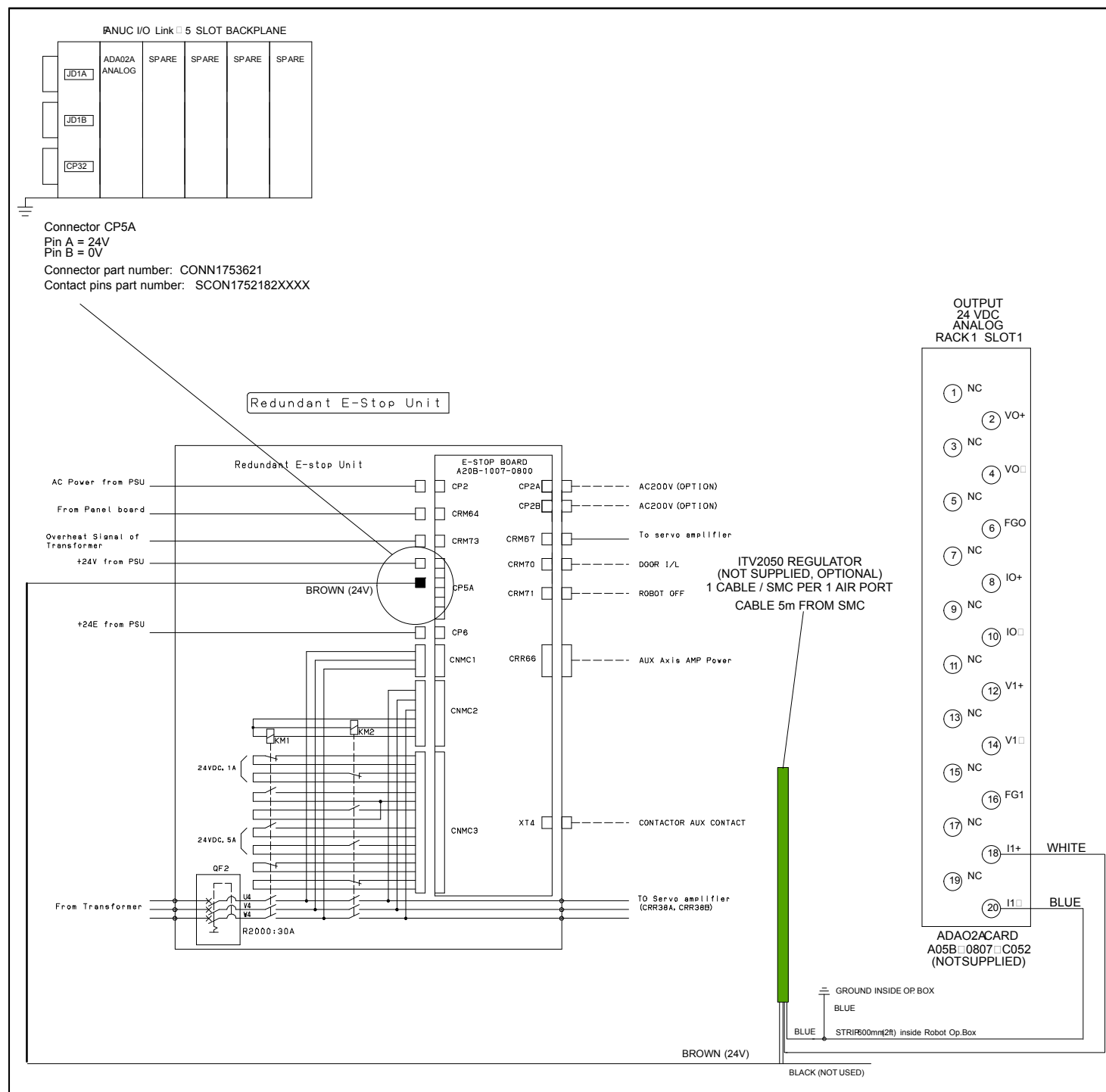
2.9 I/O SETUP

If you are using a wrist or bench type passive device, you need to connect the analog output AO 1 and 2 on the AFD2000 device cable as marked in Push Corp's AFD2000 force device manual. RemovalTool uses AO[1] and AO[2] to send voltages to the AFD device to set force specified in R[5: Programmed force]. A digital output (as specified in STARTSP.TP, STOPSP.TP) is typically used for turning the spindle or motor on and off.

If a device other than AFD2000 is used, then each analog output port needs to be connected to the corresponding external electro-pneumatic regulator (such as the SMC-ITV2050–11N3N4 series) to control the passive force device. (Examples of these commercially available devices are Push Corp's AFD72 and AFD52.) Refer to the electrical connections diagram in [Figure 2–13](#) . For example, AO[1] is connected to the pneumatic servo value of the [+] positive direction value of the device, and AO[2] to the [-] negative direction value.

Refer to the application-specific *HandlingTool Setup and Operations Manual* for more information on setting registers.

Figure 2-13. Electrical Connections Diagram



PROGRAMMING

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3.1 WRITING A PROGRAM

You can either use a RemovalTool template program or write your own.

3.1.1 Using a RemovalTool Template Program

If you are modifying a RemovalTool template teach pendant program, the order in which the macros should be called is identified in the template program. Refer to [Appendix A](#) for the template program specific to the kind of device you are using. Use [Procedure 3-1](#) to use a RemovalTool template program.

Procedure 3-1 Using a RemovalTool Template Program

Steps

1. Press SELECT.
2. Press F1, [TYPE].
3. Select TP programs.
4. Choose the teach pendant template program appropriate for your force control device:
 - *ACTVWTMP* - For use with the Active Wrist Force Control Device.
 - *PASSWTMP* - For use with the Passive Wrist Force Control Device.

Note If you want to use the wrist compensation program, the program must start with Run MRWCOMP. See below.

```
Run MRWCOMP  
LBL[X]  
Programming  
JMP LBL[X]
```

- *PASSBTMP* - For use with the Passive Bench Device.
5. Copy the template teach pendant program you have chosen into your own teach pendant program. Refer to the "Program and File Manipulation" chapter of the *FANUC America Corporation SYSTEM R-J3iC Controller HandlingTool Setup and Operations Manual*.
 6. You can now modify the copied template program as desired.

3.1.2 Writing Your Own RemovalTool Program

If you are writing your own RemovalTool teach pendant program, you must set up the macros in the correct order. Refer to the template program specific to your force control device ([Appendix A](#)), for the correct macro order.

Refer to [Chapter 2 SETTING UP REMOVALTOOL](#) for information on how to set up a RemovalTool application before you actually begin writing the program.

For more information on how to write a teach pendant program, refer to the "Planning and Creating a Program" chapter in the *HandlingTool Setup and Operations Manual* .

3.1.3 Using RemovalTool Macros

A macro command program is a separate program that contains a series of instructions to perform a task.

This section contains a listing of the standard macros that are included with RemovalTool. It also contains a listing of the teach pendant program that stands behind each RemovalTool macro. Think of a macro as a call name for each corresponding teach pendant program.

For more information on how to use macros, refer to the "General Setup" chapter in the *FANUC America Corporation SYSTEM R-J3iC Controller HandlingTool Setup and Operations Manual* .

Macro Table

[Table 3–1](#) provides detailed information about each RemovalTool macro.

Note Active refers to AFD1000 and Passive refers to AFD2000 for ADJUST_TOOL and ADJUST_UFRAME only. AFD72 and AFD52 do not have positioned feedback and cannot be used with ADJUST_TOOL and ADJUST_UFRAME. Active/Passive can be used with all device types.

Table 3-1. RemovalTool Macros

Macro	Macro Program	Device Used In	Condition/Actions
ACTIVE SLIDE POS	SLIDEPOS.MR	Active Wrist/Bench	Retrieves the slide position from the AFD controller to the robot. This value is set in data register R[16].
ADJUST TOOL WT	MRWCOMP.MR	Passive Wrist	Uses the tool weight and force entered on the RT Data SETUP screen, to adjust for forces at different tool orientations. Continuously updates AO[1] and AO[2] based on the tool orientation.
ADJUST UFRAME	ADJUFRAM.MR	Active (AFD1000) and Passive (AFD2000) Bench	Starts the wheel wear search and shifts the USER frame by the offset found. This macro should be inserted into your teach pendant program after the search start position is taught.
ADJUST UTOOL	ADJUTOOL.MR	Active Wrist	Starts the wheel wear search and applies the offset to UTOOL[1] which is based on PR[1] (Nominal UTOOL). The ADJUSTUTOOL macro should be inserted into your teach pendant program after the search start position is taught.
MOVE TO PERCH	PRGPERCH.MR	No Device Dependence	Moves the robot to the perch position. Should be set only one time for each robot system. Value is stored in PR[10].
SET AFD FORCE	CMDFORCE.MR	Active Wrist	Sends the force entered on the RTData SETUP menu to the AFD controller. This value is set in data register R[5].
STOP ACTIVE DEV	STOPAFD.MR	Active Wrist	Terminates communication with the AFD controller.
AUTO WEIGH EOAT	AUTOWGH.MR	Active Wrist	Instructs the AFD to weigh the tool.

Table 3–1. RemovalTool Macros (Cont'd)

Macro	Macro Program	Device Used In	Condition/Actions
SET TOOL WEIGHT	SETTOOLW.MR	Active Wrist	Uses the tool weight to adjust for forces at different tool orientations. Tool weight is set in data register R[10].
START ACTIVE DEV	STARTAFD.MR	Active Wrist/Bench	Initiates communication with the AFD controller.
START SPINDLE	STARTSP.MR	Active, Passive, Wrist/Bench	Turns on the spindle. The output can be changed to either a robot digital output (RDO) or a digital output (DO). The status of ON (1) or OFF (0) is contained in data register R[1]. See STARTSP.MR
STOP SPINDLE	STOPSP.MR	Active, Passive, Wrist/Bench	Turns off the spindle. The output can be changed to either a robot digital output (RDO) or a digital output (DO). The status of ON (1) or OFF (0) is contained in data register R[1]. See STOPSP.MR

**Caution**

Do not use a macro which is not designed for your force control device. Otherwise you could damage equipment.

Macro Programs

This section contains each RemovalTool macro program.

ACTIVE SLIDE POS

The corresponding program name for this macro is SLIDEPOS.MR.

SLIDEPOS.MR

```
1:  CALL GETSLPOS ;
    /POS
    /END
```

ADJUST TOOL WT

The corresponding program name for this macro is MRWCOMP.MR. MRWCOMP.MR should run once before starting production. This program will run in the background to provide passive device tool weight and gravity compensation. Pressing FUNC and ABORT ALL will abort MRWCOMP.MR.

MRWCOMP.MR

```
1:  RUN MR_WCOMP ;  
/POS  
/END
```

ADJUST UFRAME

The corresponding program name for this macro is ADJUFRAM.MR.

ADJUFRAM.MR

```
1:  IF R[4]=2,JMP LBL[100] ;  
2:  PR[8]=PR[7] ;  
3:  PR[3]=LPOS ;  
4:  R[20]=0 ;  
5:  LBL[10:SENSOR SEARCH] ;  
6:  R[20]=R[20]+1 ;  
7:  PR[8]=PR[8]+PR[2] ;  
8:  UFRAME[1]=PR[8] ;  
9:L PR[3] 100mm/sec CNT10 ;  
10: ACTIVE SLIDE POS ;  
11: IF R[16]<.6,JMP LBL[20] ;  
12: JMP LBL[30] ;  
13: LBL[20:Retract to START] ;  
14: UFRAME[1]=PR[7] ;  
15:L PR[3] 200mm/sec CNT10 ;  
16: UFRAME[1]=PR[8] ;  
17: JMP LBL[100] ;
```

ADJUST UTOOL

The corresponding program name for this macro is ADJUTOOL.MR.

ADJUTOOL.MR

```
1:  IF R[4]=3,JMP LBL[100] ;  
2:  PR[4]=PR[1] ;
```

```
3: PR[3]=LPOS      ;
4: R[20]=0         ;
5: LBL[10:SENSOR SEARCH] ;
6: R[20]=R[20]+1   ;
7: PR[4]=PR[4]+PR[2] ;
8: UTOOL[1]=PR[4] ;
9:L PR[3] 100mm/sec CNT10 ;
10: ACTIVE SLIDE POS ;
11: IF R[16]<.6,JMP LBL[20] ;
12: JMP LBL[30] ;
13: LBL[20:Retract to START] ;
14: UTOOL[1]=PR[1] ;
15:L PR[3] 200mm/sec CNT10 ;
16: UTOOL[1]=PR[4] ;
17: JMP LBL[100] ;
18: LBL[50:Sensor Alarm] ;
19: UALM[1] ;
20: PAUSE ;
21: ABORT ;
22: JMP LBL[100] ;
23: LBL[30:Check Srch Range] ;
24: IF R[20]=20,JMP LBL[50] ;
25: JMP LBL[10] ;
26: LBL[100:End of Program] ;
```

COMMAND FORCE

The corresponding program name for this macro is CMDFORCE.MR.

CMDFORCE.MR

```
1: CALL CMD_FORC ;
/POS
/END
```

DEFAULT DELAY

The corresponding program name for this macro is MRWAIT.MR.

MRWAIT.MR

```
1: CALL MRWAIT ;
/POS
/END
```

MOVE TO PERCH

The corresponding program name for this macro is PRGPERCH.MR.

PRGPERCH.MR

```
1: J PR[10] 25% FINE ;  
/POS  
/END
```

SET TOOL WEIGHT

The corresponding program name for this macro is SETTOOLW.MR.

SETTOOLW.MR

```
1: CALL SET_TOOL ;  
/POS  
/END
```

START ACTIVE DEV

The corresponding program name for this macro is STARTAFD.MR.

STARTAFD.MR

```
1: CALL STRT_AFD ;  
/POS  
/END
```

START SPINDLE

The corresponding program name for this macro is STARTSP.MR.

STARTSP.MR

```
1: IF R[25]=0,JMP LBL[11] ;  
2: RO[1]=ON ;  
3: WAIT .50(sec) ;  
4: R[25]=0 ;  
5: LBL[11] ;  
6: R[1]=1 ;
```

STOP SPINDLE

The corresponding program name for this macro is STOPSP.MR.

STOPSP.MR

```
1: DO[1]=OFF ;  
2: R[1]=0 ;  
/POS  
/END
```

3.1.4 Adjust UTOOL or UFRAME

After the search start position is taught, the ADJUTOOL or ADJUFRAM macro must be inserted in the teach pendant program. Whether you insert ADJUTOOL or ADJUFRAM depends on the kind of force control device you are using. Refer to [Table 3–1](#) if you do not know which one to use.

3.2 FORCE, TOOL WEAR, AND WHEEL WEIGHT COMPENSATION

The RemovalTool software is designed to compensate for certain inevitable situations that will arise when you run a removal operation.

3.2.1 Setting Forces

The command force in an application is specified for the nominal orientation of the tool. In order to maintain the specified force between contact points, the force value must be modified at tool orientations other than the nominal orientation.

After the command force is set, the RemovalTool software automatically takes into account the tool weight and tool orientation and maintains the command force within a specified tolerance.

There should only be minor fluctuations in this value as stated in the datasheet. The number should remain relatively stable, regardless of the slide position or device orientation.

If you are using the active wrist force control device in your program, set R[5:Programmed Forces]=value and then call the SET AFD FORCE macro. This macro communicates the value in R[5] to the force controller.

If you are using the passive wrist force control device, do the following:

1. Set up the electrical and pneumatic connection as described in [Chapter 2 SETTING UP REMOVALTOOL](#) for the dual acting passive device.

2. Configure RemovalTool software for the kind of passive device you have:
 - a. Set FP_PARAM and FN_PARAM as described in [Section 2.7.1](#) .
 - b. Set registers R[4], R[5], R[10], and R[15] as described in [Section B.1](#) . Refer to [Table B-1](#) .
3. Run MRWCOMP.MR program once to initiate tool weight compensation. Then change R[5:FORCE]=value as the process requires.

The force value will be applied automatically into the passive device as MR_WCOMP tool weight compensation program is running in the background. Therefore, you can change force values in the production program and the RemovalTool macro will automatically update the applied force.

Note To stop tool weight compensation, press FUNC and select ABORT ALL. This will stop MR_WCOMP macro from running in the background.

3.2.2 Tool Wear Compensation

Tool wear applies only to soft media such as sand disk, grinding wheels, and scotch brite wheels.

Tool wear compensation applies to all tools: active wrist, passive wrist, and passive bench mount. As the tool (abrasive wheel diameter for example) wears out, FANUC recommends that you detect it at the start of the operation by using the ADJUST UTOOL (for wrist mount) or ADJUST UFRAME (for bench mount) devices. This modifies the Tool Frame by the tool wear amount. Taught robot positions are not modified by the software.

The overall process cycle time can be reduced significantly, if the amount of tool wear is measured accurately and the tool frame is modified accordingly.

You must also teach a search start position right before calling the Adjust UTOOL or Adjust UFRAME macro. During operation, the robot will move to this search start position and then move by a distance of -2mm along the user selected slide direction, until either the stroke sensor goes from ON to OFF (as in the case of the passive wrist device) or the slide direction is at 0.4 inches (as in the case of the active wrist device) after contact is made.

When the bench mounted device is used, the User Frame is shifted along the selected slide direction until the stroke sensor goes from ON to OFF (as in the case of the passive wrist device) or the slide direction is at 0.4 inches (as in the case of the active wrist device) after contact is made. The modified User Frame is used to compensate for wheel wear and maintain the slide at its mid stroke position.

Refer to [Section 2.6](#) for more information about searching distance and direction for wheel wear.

3.2.3 Wheel Weight Compensation

Wheel weight compensation can be used with active wrist only. Wheel weight compensation can be used only with sanding or grinding media that loses its weight due to process.

If you have an active wrist force control device , wheel weight can be automatically determined by the AFD controller using the Autoweigh EOAT macro.

If you have a passive wrist force control device , no force feedback is available. ADJUST TOOL WT macro should be used to apply the programmed force, based on tool orientation and tool weight. The macro uses the setup information to estimate the tool weight based on the amount of tool wear, and provide the required compensation.

TESTING A PROGRAM AND RUNNING PRODUCTION

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4.1 TESTING A REMOVALTOOL PROGRAM

Test run RemovalTool after you have completed all setup. Refer to [Chapter 2 SETTING UP REMOVALTOOL](#) for setup information.

After you test run for the first time you might need to adjust the path or other setup data and test again. Refer to [Section 4.2](#) for some tips and guidelines about RemovalTool performance.

4.1.1 Testing a Program

Use [Procedure 4-1](#) to test a RemovalTool program.

Procedure 4-1 Testing a RemovalTool Program

Conditions

- The force control device has been installed. Refer to [Chapter 2 SETTING UP REMOVALTOOL](#)
- All personnel and unnecessary equipment are out of the workcell

Steps

1. To teach a perch position,

- a. Press SELECT.
- b. Press F1, [TYPE].
- c. Select Macro.
- d. Select PRGPERCH and press ENTER. You will see a screen similar to the following.

```
1/2  
1 : J PR[10] 5% FINE  
[End]
```

- e. Jog the robot to a perch position above the height of the highest part.

2. To record a position, press SHIFT and F4, TOUCHUP.

Note You can teach only one perch position per robot system. Make sure you teach the perch position so that it will accommodate any operation you might run in that workcell. After the perch position is taught it does not have to be re-taught.

3. Check the air supply and test the tool device before you test run the RemovalTool application.

- a. Verify that the air supply is adequate by looking at the gauge.

For both active and passive wrist force control devices, the air supply should be maintained at 100 psi (pounds per square inch).

4. To test run the application

- a. Set the REMOTE/LOCAL setup item in the System Configuration Menu to LOCAL.
- b. Set the jog speed to 25% or less.
- c. Select the template program you want to use from the SELECT menu.



Warning

The next step causes robot motion. Make sure all safety barriers are in place, all personnel are outside of the workcell, all equipment is in place, and all production conditions have been met before you continue. Otherwise, you could injure personnel or damage equipment.

- d. Move the robot to the perch position using the PRGPERCH macro program.
- e. Press the CYCLE START button on the controller to start the application program.

The robot will move to the search start position and use the Adjust Utool macro to check for wheel wear.

The application program will automatically begin running.

- f. Press the Display Status key on the Teach Pendant.

5. If you want to track the activity of the controller,

- a. Press MENU.
- b. Select ALARM.
- c. Press F1, [TYPE].
- d. Select Appl Log.

Messages will be listed on the screen.

4.1.2 Monitoring Teach Pendant Programs

You can monitor a running teach pendant program from the SELECT menu. When you monitor a running program, the program is displayed and the cursor highlights the line currently being executed.

Use [Procedure 4-2](#) to monitor a running teach pendant program.

Procedure 4-2 Monitoring a Running Teach Pendant Program

Conditions

- The teach pendant program or programs you want to monitor is (are) currently executing.

Steps

- 1.** Press SELECT.

```
Select
                                50983 BYTES FREE
No.  Program name                Comment
  1  SUB1                        PR [
  2  MAIN25                      JB [
  3  PRG7                        MR [
  4
  5
  6
```

2. Press F4, MONITOR. The program will be displayed on the screen.

If more than one program is running, you will see a screen similar to the following. The cursor will be on the line number of the instruction that is currently being executed. The cursor will move to each instruction as it is executed.

```
LONGDELAYPROGRA> LINE 1      AUTO RUNNING\\\\\\\\\\\\\\\\\\\\JOINT\\100\\%
Select\\(monitor)\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\
                TASK              STATUS
\\1\\EXEC1\\\\\\\\\\LONGDEL>\\[LONGDEL>:Running\\]
 2 EXEC2      DELAY1    [DELAY1   :Running ]
 3 EXEC3      DELAY2    [DELAY2   :Running ]

Select monitor program
  PROG     ROUT      PAUSE      ABORT
```

```

LONGDELAYPROGRA>  LINE 1      AUTO RUNNING\////////////////\JOINT\100\%
Select\ (monitor) \\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\
                TASK          STATUS
\1\EXEC1\\\\\LONGDEL>\ [LONGDEL>:Running\]
 2 EXEC2      DELAY1    [DELAY1   :Running ]
 3 EXEC3      DELAY2    [DELAY2   :Running ]

```



```

LONGDELAYPROGRAMNAME
  PROG      ROUT      PAUSE      ABORT

```

- ## 4.2 PERFORMANCE FACTORS AND GUIDELINES

As the user, you should be aware that you have control over these factors, and thus the performance of your application. This section contains some tips and guidelines that might help you to improve the performance of your application.

4.2.1 Tool Orientation

If your application uses an abrasive disc, the tool should be rotated, such that only the outer edge of the disc makes contact with the part surface. Otherwise, the abrasive plate can get damaged easily.

Keep in mind that the command force is the force applied along the direction of the slide. If this slide is not perpendicular to the contact surface, only one component (along the direction of the slide) of the actual applied force is equal to the command force.

It is highly recommended that you use the 6 point method of teaching the UTOOL. In order for the wrist-mounted force device to operate properly, the positive slide direction must be along the -x direction of the UTOOL.

4.2.2 Robot Speed

Robot speed depends mainly on the abrasive media, spindle speed, and part surface. However, speeds above 150 mm/sec are not recommended. They might cause the weight compensation to be ineffective. When you test run your application or run production, FANUC America Corporation recommends always setting the override to 100.

4.2.3 Spindle Speed

FANUC America Corporation currently offers spindles that have only one speed. However, the RemovalTool software has the capability to implement multiple speed spindles. This requires appropriate I/O wiring and additional programming within the user application program. Contact FANUC America Corporation before using spindles with multiple speeds or any other custom spindles.

4.2.4 Tool Media

RemovalTool software handles only wheel, cup, and disc media. However, it is possible to implement other tool media in RemovalTool. Contact FANUC America Corporation before you use any other tool media.

Note FANUC America Corporation advises you to select the abrasive plate carefully, based on process experience, performance requirement, and the life expectancy of the abrasive plate.

4.2.5 Path Teaching Guidelines

The way in which you teach the robot positions has a considerable impact on the process performance. Make sure that the contact time between part surface and tool media is kept consistent throughout the robot path. This ensures that the removal process is consistent.

Since the tool is expected to operate at its mid-stroke, the positions at the beginning of the path must be slightly away from the part surface and gradually touch the surface along its path. Similarly, the

robot must retract away gradually to make sure that the contact time of the tool with the part surface is consistent. Otherwise more material might be removed at the start and end point of the robot path.

It is also important that the tool orientation is changed gradually whenever possible. This allows the robot to compensate effectively for the tool weight and maintain the command force at the contact point. This might not be possible while moving the tool around sharp corners. However, you can slow down the robot speed and reduce the spindle speed to get better results. Circular motion type is recommended at corner positions.

4.2.6 ESTOP or PAUSE

RemovalTool retracts the tool automatically and stops the spindle from spinning whenever the robot is in an EMERGENCY STOP or PAUSE condition. This separates the tool from the work piece and protects both the tool and the part. This is achieved from the ESTOP-CH program.

4.3 EXECUTING MACRO COMMANDS

Refer to Section for information on RemovalTool macros. Refer to the application-specific *HandlingTool Setup and Operations Manual* for information on setting up and executing macros and on the macro command program instruction.

4.4 PRODUCTION OPERATION

Refer to the application-specific *HandlingTool Setup and Operations Manual* for more information on production operation setup and execution.

ERROR CODES AND TROUBLESHOOTING

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5.1 ERROR CODES

Error codes appear on the alarm log.

Refer to the *Error Code Manual* for a complete listing of error codes.

5.2 TROUBLESHOOTING

This section contains information that will help you solve problems you might have while running RemovalTool.

[Table 5–1](#) includes common symptoms and possible solutions to the problems.

Table 5–1. Troubleshooting Symptoms and Solutions

Symptom	Possible Cause	Possible Solution
The force control device is searching in the wrong direction.	The UFRAME or UTOOL is set incorrectly.	Re-teach the UFRAME with the three point method, or UTOOL with the six point method by following the procedures in the application-specific <i>HandlingTool Setup and Operations Manual</i> .
The slide does not go to mid stroke during the application.	The UTOOL might not be adjusted properly.	Make sure the Adjust Utool macro has been inserted in the proper place in the user teach pendant program.
The tool does not extend.	The proper force is not being applied.	For the active force device, make sure the tool weight is set properly, the communication cable is connected, the AFD communication LED is flashing, the air hose is connected and the air pressure is being maintained at 80 to 100 psi. For the passive device, make sure the air pressure is at least 60 psi.
The search ends up in an alarm or error condition.	The search range set in the ADJUTOOL teach pendant program might have been exceeded.	Teach the search start position so that it is just touching the part surface and the tool is fully extended.

Table 5–1. Troubleshooting Symptoms and Solutions (Cont'd)

Symptom	Possible Cause	Possible Solution
The robot does not seem to move along the taught path.	The proper UTOOL or UFRAME was not set on the RT Data screen.	Make sure the program has the proper UTOOL or UFRAME.
The robot does not search in the proper direction while running Adjust Utool.	UTOOL or UFRAME was taught incorrectly.	Make sure UTOOL is correct by jogging in the Tool Coordinate Frame.
	The <i>slide direction</i> is incorrect.	Check the slide direction.
Spindle does not turn ON or OFF.	The Start Spindle/Stop Spindle macros are not used in the tp program.	Compare your Teach Pendant program with the corresponding template program to make sure you are using the correct macro.
	The specified I/O for spindle ON/OFF is incorrect.	Make sure the I/O spindle ON and OFF macros are set up correctly.
	The spindle does not have air or power.	Check the power and air supply. This can be done manually by turning on the appropriate I/O to start the spindle.
Spindle turns on too late.	The spindle needs some time to reach its r.p.m.	Turn on the spindle before the last approach point.
Spindle stalls.	Too much force is being used.	Decrease the amount of force used.
Slide does not engage.	The slide does not have air.	Make sure the slide is receiving air.
	Active Force Device controller communication is not met (active device only).	Make sure the AFD communication line in the controller is working properly. If it is, the communication light on the AFD controller will be flashing.

Table 5–1. Troubleshooting Symptoms and Solutions (Cont'd)

Symptom	Possible Cause	Possible Solution
The force device does not apply the correct force to the part.	The proper <i>force</i> was not specified.	Make sure the desired force has been set correctly.
	The proper <i>tool weight</i> was not specified.	Make sure the tool weight has been set correctly. For the AFD, the autoweight function can be used.
	For the passive device, the standard <i>scale factor</i> might have changed.	Check R[21] to see if the pneumatic standard scale factor is = 75. Make sure the control valves are functioning properly.
	For the active device, the AFD controller communication is not working.	Clear the communication port for the AFD controller.
	The SET AFD FORCE and/or SET TOOL WT macro(s) are not being used.	Make sure the SET AFD FORCE macro has been inserted in your teach pendant program at the correct location. Refer to the ACTVTEMP.TP program in Appendix A
Robot takes off more material at the beginning and end of the part.	The <i>approach points</i> to the part might be too steep.	Make sure the approach points achieve the desired motions.
Robot takes off too much material.	The applied force is too high.	Lower the applied force.
	The robot speed is too slow.	Increase the robot speed.
	The spindle speed is too fast (for variable speed spindles only).	Decrease the spindle speed.
	The media being used is incorrect for the application.	Make sure the proper media is being used.
"Robot is not at perch" message is given at the start of the cycle, even though the robot seems to be at the perch position.	The perch position was not taught with the nominal UTOOL.	Re-teach the perch position using the nominal UTOOL.

REMOVALTOOL PROGRAMS

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 A.2.3 Passive Bench Mount Force Control Device A-5

A.1 MR TEACH PROGRAM

The MR TEACH program is the program that is responsible for extending the active wrist, passive wrist, or bench mounted slide. It also sets the values of certain registers and analog outputs, and initiates certain RemovalTool macros.

```

1:  IF R[4]=1, JMP LBL[10] ;
2:  IF R[4]=2, JMP LBL[20] ;
3:  !Passive Bench Device ;
4:  R[6]=R[5]*R[21] ;
5:  AO[1]=R[6] ;
6:  JMP LBL[100] ;
7:  ;
8:  LBL[10:Active Wrist] ;
9:  START ACTIVE DEV ;
10: SET AFD FORCE ;
11: JMP LBL[100] ;
12: ;
13: LBL[20:Passive Wrist]

```

A.2 TEMPLATE PROGRAMS

The following template programs can be used as they are, or modified to fit your specific removal application needs.

There are three template programs included in this section. Use the program that corresponds to the force control device type you have.

- If you have an active wrist force control device, use the program in [Section A.2.1](#)
- If you have a passive wrist force control device, use the program in [Section A.2.2](#)
- If you have a passive bench mounted device, use the program in [Section A.2.3](#)

A.2.1 Active Wrist Force Control Device

The following template program should be used if you have an active wrist force control device mounted on your robot.

Figure A–1. Active Wrist Template Program

```

1:  !***** ;
2:  !* TEMPLATE PROGRAM FOR ACTIVE * ;
3:  !* FORCE DEVICE. * ;
4:  !***** ;
5:  MOVE TO PERCH ;
6:  START ACTIVE DEV ;

```

```

7: !Tool Weight ;
8: AUTO WEIGH EOAT ;
9: !SET Command force (lb) in R[5] ;
10: R[5]=5 ;
11: SET AFD FORCE ;
12: ;
13: !***** ;
14: !*Teach Search Start Pos. * ;
15: !***** ;
16: ;
17: L P[1] 100mm/sec CNT10 ;
18: ADJUST UTOOL ;
19: ;
20: LBL[10:START] ;
21: ;
22: !***** ;
23: !*Teach Approach Positions * ;
24: !***** ;
25: ;
26: L P[2] 100mm/sec CNT100 ;
27: L P[3] 100mm/sec CNT10 ;
28: ;
29: START SPINDLE ;
30: ;
31: !***** ;
32: !*Teach part Positions. * ;
33: !***** ;
34: ;
35: !***** ;
36: !*Teach Departure Positions. * ;
37: !***** ;
38: ;
39: STOP SPINDLE ;
40: MOVE TO PERCH ;
41: STOP ACTIVE DEV ;
42: ;
43: END ;

```

A.2.2 Passive Wrist Force Control Device

The following template program should be used if you have a passive wrist force control device mounted on your robot.

Figure A-2. Passive Wrist Template Program

```

1:  !***** ;
2:  !* TEMPLATE PROGRAM FOR PASSIVE* ;
3:  !* WRIST FORCE DEVICE.          * ;
4:  !***** ;
5:  ;
6:  !***** ;
7:  !*Teach Perch @ PR(10)          * ;
8:  !***** ;
9:  MOVE TO PERCH      ;
10:  ADJUST TOOL WT    ;
11:  ;
12:  !***** ;
13:  !Teach Search Start Pos.        * ;
14:  !***** ;
15: L P[1] 100mm/sec FINE      ;
16:  ;
17:  ADJUST UTOOL      ;
18:  ;
19:  LBL[10:START]      ;
20:  ;
21:  !***** ;
22:  !*Teach Approach Positions      * ;
23:  !***** ;
24: J P[2] 100% CNT20      ;
25: J P[3] 50% CNT20      ;
26:  START SPINDLE      ;
27: J P[4] 50% CNT20      ;
28:  ;
29:  !***** ;
30:  !*Teach part Positions.          * ;
31:  !***** ;
32: L P[5] 100mm/sec CNT100    ;
33: L P[6] 100mm/sec CNT100    ;
34:  ;
35:  !***** ;
36:  !*Teach Departure Positions.      * ;
37:  !***** ;
38: L P[7] 250mm/sec CNT100    ;
39: J P[8] 100% CNT50      ;
40:  ;
41:  MOVE TO PERCH      ;
42:  STOP SPINDLE      ;
43:  ;
44:  END ;

```

A.2.3 Passive Bench Mount Force Control Device

The following template program should be used if you have a passive bench mount force control device mounted on your robot.

Figure A-3. Passive Bench Mount Template Program

```

1:  !***** ;
2:  !* TEMPLATE PROGRAM FOR PASSIVE* ;
3:  !* BENCH DEVICE.                * ;
4:  !***** ;
5:  ;
6:  !***** ;
7:  !*Teach Perch @ PR(10)          * ;
8:  !***** ;
9:  MOVE TO PERCH      ;
10: ;
11: !Set command force via  ;
12: !AO[1] : extend ;
13: !AO[2] : retract ;
14: ;
15: !***** ;
16: !Teach Search Start Pos.      * ;
17: !THIS POSITION MUST BE        * ;
18: !PERPENDICULAR TO THE CENTER * ;
19: !OF THE WHEEL. ;
20: !***** ;
21: L P[1] 100mm/sec FINE      ;
22: ;
23: ADJUST UFRAME      ;
24: ;
25: LBL[10:START] ;
26: ;
27: !***** ;
28: !*Teach Approach Positions    * ;
29: !***** ;
30: J P[2] 100% CNT20      ;
31: J P[3] 50% CNT20      ;
32: START SPINDLE      ;
33: L P[4] 100mm/sec CNT100  ;
34: ;
35: !***** ;
36: !*Teach part Positions.      * ;
37: !***** ;
38: ;
39: L P[5] 100mm/sec CNT100  ;
40: L P[6] 100mm/sec CNT100  ;
41: ;

```

```
42:  !***** ;
43:  !*Teach Departure Positions.  * ;
44:  !***** ;
45:L P[7] 250mm/sec CNT100 ;
46:J P[8] 100% CNT50 ;
47:  ;
48:  MOVE TO PERCH ;
49:  STOP SPINDLE ;
50:  ;
51:  END ;
```

REMOVALTOOL DATA

Contents

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B.1 REGISTER ASSIGNMENTS

This section includes register assignments for RemovalTool.

Refer to [Table B-1](#) for RemovalTool Data Register Assignments.

Table B-1. Data Register Assignments

Instruction	Description
R[1:Spindle status]	Reserved for FANUC America Corporation
R[2:Product-id]	Reserved for FANUC America Corporation
R[3:Reserved]	Reserved for FANUC America Corporation
R[4:Force Device]	The type of force control device or wrist attached to your robot. 1=Active Wrist 2=Passive Wrist 3=Passive Bench Mount
R[5:Programmed force (lb)]	The amount of force to be applied in pound force. [lbf] Conversion metric $\Rightarrow$ English: $X[lbf] = F[N] / 4.5$
R[6:Slide Dir Comp]	The positive slide direction of the wrist mounted force device. 0=Slide in -x direction 1=Slide in +z direction
R[7: Reverse Force]	Reserved for FANUC America Corporation
R[8:Tool medium]	Reserved for FANUC America Corporation
R[9:Wheel radius]	Reserved for FANUC America Corporation
R[10:Tool weight]	The weight of the tool (in pounds). This includes the motor, tool media, slide, and any other device used to mount the tool to the force control device.
R[11:Spindle speed]	Reserved for FANUC America Corporation
R[12: Robot pos source]	Reserved for FANUC America Corporation
R[13:]	Reserved for FANUC America Corporation
R[14:]	Reserved for FANUC America Corporation
R[15:Slide direction]	The direction that the tool slides. This information is used during production to adjust the Tool Frame and weight compensation. = 1 if the slide is along the UTOOL +x direction = 2 if the slide is along the UTOOL +y direction = 3 if the slide is along the UTOOL +z direction
R[16:AFD Slide Pos]	Reserved for FANUC America Corporation
R[17:]	
R[18:Reg_DELAY]	Reserved for FANUC America Corporation

Table B-1. Data Register Assignments (Cont'd)

Instruction	Description
R[19:]	
R[20:Search Cycle Cnt]	This register contains the number of search attempts after running the ADJUST UTOOL or ADJUST UFRAME macros.
R[21:Scale Factor]	Reserved for FANUC America Corporation
R[22:AFD Error Number]	This register contains the identification number of the error received.
R[23:]	Reserved for FANUC America Corporation
R[24:]	Reserved for FANUC America Corporation
R[25:]	Reserved for FANUC America Corporation
R[26:]	Reserved for FANUC America Corporation
R[27:]	Reserved for FANUC America Corporation
R[28:]	Reserved for FANUC America Corporation
R[29: Compensation Amt]	Reserved for FANUC America Corporation
R[30:CYC BEFORE UT ADJ]	The number of cycles before the robot adjusts the Utool.
R[31:TOTAL CYCLES]	The total number of times the application is run in production.
R[32:Current Cycle]	
R[33:Abort Wear Prog]	Reserved for FANUC America Corporation
R[34:Abort\mr_wcomp]	Reserved for FANUC America Corporation
R[35:Heartbeat DOUT]	Reserved for FANUC America Corporation

B.2 POSITION REGISTER ASSIGNMENTS

This section includes position register assignments for RemovalTool.

Refer to [Table B-2](#) for RemovalTool Position Register Assignments.

Table B-2. Position Register Assignments

Instruction	Description
PR[1:Nominal UTOOL]	The original positions to which all points must be set, without any offsets. The macro SET NOM UTOOL sets this value.
PR[2:Incr search dist]	The incremental amount the robot moves to detect wheel wear. Refer to Section 2.6
PR[3:Reserved]	

Table B-2. Position Register Assignments (Cont'd)

Instruction	Description
PR[4:Effective UTOOL]	The offset needed after wheel wear plus 1/2 the slide length. The macro SET EFFECT UTOOL sets this value.
PR[5:]	
PR[6:Reserved]	
PR[7:Nominal UFRAME]	The Nominal UFRAME is stored here.
PR[8:Effective UFRAME]	The Effective UFRAME (for the bench mounted device) is stored here.
PR[9:]	
PR[10:PERCH POS]	The perch or home position that was set up for the robot system. The macro MOVE TO PERCH uses this value.

B.3 I/O ASSIGNMENTS

The recommended I/O assignments are:

Table B-3. Recommended I/O Assignments

DO[1]	For turning the spindle ON/OFF
AO[1]	For extending the passive device (wrist or bench) positive “+”
AO[2]	For retracting the passive device (wrist or bench) negative “-”

Glossary

A

abort

Abnormal termination of a computer program caused by hardware or software malfunction or operator cancellation.

absolute pulse code system

A positional information system for servomotors that relies on battery-backed RAM to store encoder pulse counts when the robot is turned off. This system is calibrated when it is turned on.

A/D value

An analog to digital-value. Converts a multilevel analog electrical system pattern into a digital bit.

AI

Analog input.

AO

Analog output.

alarm

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

algorithm

A fixed step-by-step procedure for accomplishing a given result.

alphanumeric

Data that are both alphabetical and numeric.

AMPS

Amperage amount.

analog

The representation of numerical quantities by measurable quantities such as length, voltage or resistance. Also refers to analog type I/O blocks and distinguishes them from discrete I/O blocks. Numerical data that can vary continuously, for example, voltage levels that can vary within the range of -10 to +10 volts.

AND

An operation that places two contacts or groups of contacts in series. All contacts in series control the resulting status and also mathematical operator.

ANSI

American National Standard Institute, the U.S. government organization with responsibility for the development and announcement of technical data standards.

APC

See absolute pulse code system.

APC motor

See servomotor.

application program

The set of instructions that defines the specific intended tasks of robots and robot systems to make them reprogrammable and multifunctional. You can initiate and change these programs.

arm

A robot component consisting of an interconnecting set of links and powered joints that move and support the wrist socket and end effector.

articulated arm

A robot arm constructed to simulate the human arm, consisting of a series of rotary motions and joints, each powered by a motor.

ASCII

Abbreviation for American Standard Code for Information Interchange. An 8-level code (7 bits plus 1 parity bit) commonly used for the exchange of data.

automatic mode

The robot state in which automatic operation can be initiated.

automatic operation

The time during which robots are performing programmed tasks through unattended program execution.

axis

1. A straight line about which a robot joint rotates or moves. 2. One of the reference lines or a coordinate system. 3. A single joint on the robot arm.

B**backplane**

A group of connectors mounted at the back of a controller rack to which printed circuit boards are mated.

BAR

A unit of pressure equal to 100,000 pascals.

barrier

A means of physically separating persons from the restricted work envelope; any physical boundary to a hazard or electrical device/component.

battery low alarm

A programmable value (in engineering units) against which the analog input signal automatically is compared on Genius I/O blocks. A fault is indicated if the input value is equal to or less than the low alarm value.

baud

A unit of transmission speed equal to the number of code elements (bits) per second.

big-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

binary

A numbering system that uses only 0 and 1.

bit

Contraction of binary digit. 1. The smallest unit of information in the binary numbering system, represented by a 0 or 1. 2. The smallest division of a programmable controller word.

bps

Bits per second.

buffer

A storage area in the computer where data is held temporarily until the computer can process it.

bus

A channel along which data can be sent.

bus controller

A Genius bus interface board for a programmable controller.

bus scan

One complete communications cycle on the serial bus.

Bus Switching Module

A device that switches a block cluster to one bus or the other of a dual bus.

byte

A sequence of binary digits that can be used to store a value from 0 to 255 and usually operated upon as a unit. Consists of eight bits used to store two numeric or one alpha character.

C**calibration**

The process whereby the joint angle of each axis is calculated from a known reference point.

Cartesian coordinate system

A coordinate system whose axes (x, y, and z) are three intersecting perpendicular straight lines. The origin is the intersection of the axes.

Cartesian coordinates

A set of three numbers that defines the location of a point within a rectilinear coordinate system and consisting of three perpendicular axes (x, y, z).

cathode ray tube

A device, like a television set, for displaying information.

central processing unit

The main computer component that is made up of a control section and an arithmetic-logic section. The other basic units of a computer system are input/output units and primary storage.

channel

The device along which data flow between the input/output units of a computer and primary storage.

character

One of a set of elements that can be arranged in ordered groups to express information. Each character has two forms: 1. a man-intelligible form, the graphic, including the decimal digits 0-9, the letters A-Z, punctuation marks, and other formatting and control symbols; 2. a computer intelligible form, the code, consisting of a group of binary digits (bits).

circular

A MOTYPE option in which the robot tool center point moves in an arc defined by three points. These points can be positions or path nodes.

clear

To replace information in a storage unit by zero (or blank, in some machines).

closed loop

A control system that uses feedback. An open loop control system does not use feedback.

C-MOS RAM

Complementary metal-oxide semiconductor random-access memory. A read/write memory in which the basic memory cell is a pair of MOS (metal-oxide semiconductor) transistors. It is an implementation of S-RAM that has very low power consumption, but might be less dense than other S-RAM implementations.

coaxial cable

A transmission line in which one conductor is centered inside and insulated from an outer metal tube that serves as the second conductor. Also known as coax, coaxial line, coaxial transmission line, concentric cable, concentric line, concentric transmission line.

component

An inclusive term used to identify a raw material, ingredient, part or subassembly that goes into a higher level of assembly, compound or other item.

computer

A device capable of accepting information, applying prescribed processes to the information, and supplying the results of these processes.

configuration

The joint positions of a robot and turn number of wrist that describe the robot at a specified position. Configuration is designated by a STRING value and is included in positional data.

continuous path

A trajectory control system that enables the robot arm to move at a constant tip velocity through a series of predefined locations. A rounding effect of the path is required as the tip tries to pass through these locations.

continuous process control

The use of transducers (sensors) to monitor a process and make automatic changes in operations through the design of appropriate feedback control loops. While such devices historically have been mechanical or electromechanical, microcomputers and centralized control is now used, as well.

continuous production

A production system in which the productive equipment is organized and sequenced according to the steps involved to produce the product. Denotes that material flow is continuous during the production process. The routing of the jobs is fixed and set-ups are seldom changed.

controlled stop

A controlled stop controls robot deceleration until it stops. When a safety stop input such as a safety fence signal is opened, the robot decelerates in a controlled manner and then stops. After the robot stops, the Motor Control Contactor opens and drive power is removed.

controller

A hardware unit that contains the power supply, operator controls, control circuitry, and memory that directs the operation and motion of the robot and communications with external devices. See control unit.

controller memory

A medium in which data are retained. Primary storage refers to the internal area where the data and program instructions are stored for active use, as opposed to auxiliary or external storage (magnetic tape, disk, diskette, and so forth.)

control, open-loop

An operation where the computer applies control directly to the process without manual intervention.

control unit

The portion of a computer that directs the automatic operation of the computer, interprets computer instructions, and initiates the proper signals to the other computer circuits to execute instructions.

coordinate system

See Cartesian coordinate system.

CPU

See central processing unit.

CRT

See cathode ray tube.

cps (viscosity)

Centipoises per second.

CRT/KB

Cathode ray tube/keyboard. An optional interface device for the robot system. The CRT/KB is used for some robot operations and for entering programs. It can be a remote device that attaches to the robot via a cable.

cycle

1. A sequence of operations that is repeated regularly. The time it takes for one such sequence to occur. 2. The interval of time during which a system or process, such as seasonal demand or a manufacturing operation, periodically returns to similar initial conditions. 3. The interval of time during which an event or set of events is completed. In production control, a cycle is the length of time between the release of a manufacturing order and shipment to the customer or inventory.

cycle time

1. In industrial engineering, the time between completion of two discrete units of production. 2. In materials management, the length of time from when material enters a production facility until it exits. See throughput.

cursor

An indicator on a teach pendant or CRT display screen at which command entry or editing occurs. The indicator can be a highlighted field or an arrow (> or ^).

cylindrical

Type of work envelope that has two linear major axes and one rotational major axis. Robotic device that has a predominantly cylindrical work envelope due to its design. Typically has fewer than 6 joints and typically has only 1 linear axis.

D**D/A converter**

A digital-to-analog converter. A device that transforms digital data into analog data.

D/A value

A digital-to-analog value. Converts a digital bit pattern into a multilevel analog electrical system.

daisy chain

A means of connecting devices (readers, printers, etc.) to a central processor by party-line input/output buses that join these devices by male and female connectors. The last female connector is shorted by a suitable line termination.

daisy chain configuration

A communications link formed by daisy chain connection of twisted pair wire.

data

A collection of facts, numeric and alphabetical characters, or any representation of information that is suitable for communication and processing.

data base

A data file philosophy designed to establish the independence of computer program from data files. Redundancy is minimized and data elements can be added to, or deleted from, the file designs without changing the existing computer programs.

DC

Abbreviation for direct current.

DEADMAN switch

A control switch on the teach pendant that is used to enable servo power. Pressing the DEADMAN switch while the teach pendant is on activates servo power and releases the robot brakes; releasing the switch deactivates servo power and applies the robot brakes.

debugging

The process of detecting, locating and removing mistakes from a computer program, or manufacturing control system. See diagnostic routine.

deceleration tolerance

The specification of the percentage of deceleration that must be completed before a motion is considered finished and another motion can begin.

default

The value, display, function or program automatically selected if you have not specified a choice.

deviation

Usually, the absolute difference between a number and the mean of a set of numbers, or between a forecast value and the actual data.

device

Any type of control hardware, such as an emergency-stop button, selector switch, control pendant, relay, solenoid valve, or sensor.

diagnostic routine

A test program used to detect and identify hardware/software malfunctions in the controller and its associated I/O equipment. See debugging.

diagnostics

Information that permits the identification and evaluation of robot and peripheral device conditions.

digital

A description of any data that is expressed in numerical format. Also, having the states On and Off only.

digital control

The use of a digital computer to perform processing and control tasks in a manner that is more accurate and less expensive than an analog control system.

digital signal

A single point control signal sent to or from the controller. The signal represents one of two states: ON (TRUE, 1. or OFF (FALSE, 0).

directory

A listing of the files stored on a device.

discrete

Consisting of individual, distinct entities such as bits, characters, circuits, or circuit components. Also refers to ON/OFF type I/O blocks.

disk

A secondary memory device in which information is stored on a magnetically sensitive, rotating disk.

disk memory

A non-programmable, bulk-storage, random-access memory consisting of a magnetized coating on one or both sides of a rotating thin circular plate.

drive power

The energy source or sources for the robot servomotors that produce motion.

DRAM

Dynamic Random Access Memory. A read/write memory in which the basic memory cell is a capacitor. DRAM (or D-RAM) tends to have a higher density than SRAM (or S-RAM). Due to the support circuitry required, and power consumption needs, it is generally impractical to use. A battery can be used to retain the content upon loss of power.

E**edit**

1. A software mode that allows creation or alteration of a program. 2. To modify the form or format of data, for example, to insert or delete characters.

emergency stop

The operation of a circuit using hardware-based components that overrides all other robot controls, removes drive power from the actuators, and causes all moving parts of to stop. The operator panel and teach pendant are each equipped with EMERGENCY STOP buttons.

enabling device

A manually operated device that, when continuously activated, permits motion. Releasing the device stops the motion of the robot and associated equipment that might present a hazard.

encoder

1. A device within the robot that sends the controller information about where the robot is. 2. A transducer used to convert position data into electrical signals. The robot system uses an incremental optical encoder to provide position feedback for each joint. Velocity data is computed from the encoder signals and used as an additional feedback signal to assure servo stability.

end effector

An accessory device or tool specifically designed for attachment to the robot wrist or tool mounting plate to enable the robot to perform its intended tasks. Examples include gripper, spot weld gun, arc weld gun, spray paint gun, etc.

end-of-arm tooling

Any of a number of tools, such as welding guns, torches, bells, paint spraying devices, attached to the faceplate of the robot wrist. Also called end effector or EOAT.

engineering units

Units of measure as applied to a process variable, for example, psi, Degrees F., etc.

envelope, maximum

The volume of space encompassing the maximum designed movements of all robot parts including the end effector, workpiece, and attachments.

EOAT

See end of arm tooling, tool.

EPROM

Erasable Programmable Read Only Memory. Semiconductor memory that can be erased and reprogrammed. A non-volatile storage memory.

error

The difference in value between actual response and desired response in the performance of a controlled machine, system or process. Alarm=Error.

error message

A numbered message, displayed on the CRT/KB and teach pendant, that indicates a system problem or warns of a potential problem.

Ethernet

A Local Area Network (LAN) bus-oriented, hardware technology that is used to connect computers, printers, terminal concentrators (servers), and many other devices together. It consists of a master cable and connection devices at each machine on the cable that allow the various devices to "talk" to each other. Software that can access the Ethernet and cooperate with machines connected to the cable is necessary. Ethernets come in varieties such as baseband and broadband and can run on different media, such as coax, twisted pair and fiber. Ethernet is a trademark of Xerox Corporation.

execute

To perform a specific operation, such as one that would be accomplished through processing one statement or command, a series of statements or commands, or a complete program or command procedure.

extended axis

An optional, servo-controlled axis that provides extended reach capability for a robot, including in-booth rail, single- or double-link arm, also used to control motion of positioning devices.

F**faceplate**

The tool mounting plate of the robot.

feedback

1. The signal or data fed back to a commanding unit from a controlled machine or process to denote its response to the command signal. The signal representing the difference between actual response and desired response that is used by the commanding unit to improve performance of the controlled machine or process. 2. The flow of information back into the control system so that actual performance can be compared with planned performance, for instance in a servo system.

field

A specified area of a record used for a particular category of data. 2. A group of related items that occupy the same space on a CRT/KB screen or teach pendant LCD screen. Field name is the name of the field; field items are the members of the group.

field devices

User-supplied devices that provide information to the PLC (inputs: push buttons, limit switches, relay contacts, and so forth) or perform PLC tasks (outputs: motor starters, solenoids, indicator lights, and so forth.)

file

1. An organized collection of records that can be stored or retrieved by name. 2. The storage device on which these records are kept, such as bubble memory or disk.

filter

A device to suppress interference that would appear as noise.

Flash File Storage

A portion of FROM memory that functions as a separate storage device. Any file can be stored on the FROM disk.

Flash ROM

Flash Read Only Memory. Flash ROM is not battery-backed memory but it is non-volatile. All data in Flash ROM is saved even after you turn off and turn on the robot.

flow chart

A systems analysis tool to graphically show a procedure in which symbols are used to represent operations, data, flow, and equipment. See block diagram, process chart.

flow control

A specific production control system that is based primarily on setting production rates and feeding work into production to meet the planned rates, then following it through production to make sure that it is moving. This concept is most successful in repetitive production.

format

To set up or prepare a memory card or floppy disk (not supported with version 7.20 and later) so it can be used to store data in a specific system.

FR

See Flash ROM.

F-ROM

See Flash ROM.

FROM disk

See Flash ROM.

G

general override stat

A percentage value that governs the maximum robot jog speed and program run speed.

Genius I/O bus

The serial bus that provides communications between blocks, controllers, and other devices in the system especially with respect to GE FANUC Genius I/O.

gripper

The "hand" of a robot that picks up, holds and releases the part or object being handled. Sometimes referred to as a manipulator. See EOAT, tool.

group signal

An input/output signal that has a variable number of digital signals, recognized and taken as a group.

gun

See applicator.

H

Hand Model.

Used in Interference Checking, the Hand Model is the set of virtual model elements (spheres and cylinders) that are used to represent the location and shape of the end of arm tooling with respect to the robot's faceplate.

hardware

1. In data processing, the mechanical, magnetic, electrical and electronic devices of which a computer, controller, robot, or panel is built. 2. In manufacturing, relatively standard items such as nuts, bolts, washers, clips, and so forth.

hard-wire

To connect electric components with solid metallic wires.

hard-wired

1. Having a fixed wired program or control system built in by the manufacturer and not subject to change by programming. 2. Interconnection of electrical and electronic devices directly through physical wiring.

hazardous motion

Unintended or unexpected robot motion that can cause injury.

hexadecimal

A numbering system having 16 as the base and represented by the digits 0 through 9, and A through F.

hold

A smoothly decelerated stopping of all robot movement and a pause of program execution. Power is maintained on the robot and program execution generally can be resumed from a hold.

HTML.

Hypertext Markup Language. A markup language that is used to create hypertext and hypermedia documents incorporating text, graphics, sound, video, and hyperlinks.

http.

Hypertext transfer protocol. The protocol used to transfer HTML files between web servers.

I**impedance**

A measure of the total opposition to current flow in an electrical circuit.

incremental encoder system

A positional information system for servomotors that requires calibrating the robot by moving it to a known reference position (indicated by limit switches) each time the robot is turned on or calibration is lost due to an error condition.

index

An integer used to specify the location of information within a table or program.

index register

A memory device containing an index.

industrial robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions in order to perform a variety of tasks.

industrial robot system

A system that includes industrial robots, end effectors, any equipment devices and sensors required for the robot to perform its tasks, as well as communication interfaces for interlocking, sequencing, or monitoring the robot.

information

The meaning derived from data that have been arranged and displayed in a way that they relate to that which is already known. See data.

initialize

1. Setting all variable areas of a computer program or routine to their desired initial status, generally done the first time the code is executed during each run. 2. A program or hardware circuit that returns a program a system, or hardware device to an original state. See startup, initial.

input

The data supplied from an external device to a computer for processing. The device used to accomplish this transfer of data.

input device

A device such as a terminal keyboard that, through mechanical or electrical action, converts data from the form in which it has been received into electronic signals that can be interpreted by the CPU or programmable controller. Examples are limit switches, push buttons, pressure switches, digital encoders, and analog devices.

input processing time

The time required for input data to reach the microprocessor.

input/output

Information or signals transferred between devices, discreet electrical signals for external control.

input/output control

A technique for controlling capacity where the actual output from a work center is compared with the planned output developed by CRP. The input is also monitored to see if it corresponds with plans so that work centers will not be expected to generate output when jobs are not available to work on.

integrated circuit

A solid-state micro-circuit contained entirely within a chip of semiconductor material, generally silicon. Also called chip.

interactive

Refers to applications where you communicate with a computer program via a terminal by entering data and receiving responses from the computer.

interface

1. A concept that involves the specifications of the inter-connection between two equipments having different functions. 2. To connects a PLC with the application device, communications channel, and peripherals through various modules and cables. 3. The method or equipment used to communicate between devices.

interference zone

An area that falls within the work envelope of a robot, in which there is the potential for the robot motion to coincide with the motion of another robot or machine, and for a collision to occur.

interlock

An arrangement whereby the operation of one control or mechanism brings about, or prevents, the operations of another.

interrupt

A break in the normal flow of a system or program that occurs in a way that the flow can be resumed from that point at a later time. Interrupts are initiated by two types of signals: 1. signals originating within the computer system to synchronize the operation of the computer system with the outside

world; 2. signals originating exterior to the computer system to synchronize the operation of the computer system with the outside world.

I/O

Abbreviation for input/output or input/output control.

I/O block

A microprocessor-based, configurable, rugged solid state device to which field I/O devices are attached.

I/O electrical isolation

A method of separating field wiring from logic level circuitry. This is typically done through optical isolation devices.

I/O module

A printed circuit assembly that is the interface between user devices and the Series Six PLC.

I/O scan

A method by which the CPU monitors all inputs and controls all outputs within a prescribed time. A period during which each device on the bus is given a turn to send information and listen to all of the broadcast data on the bus.

ISO

The International Standards Organization that establishes the ISO interface standards.

isolation

1. The ability of a logic circuit having more than one inputs to ensure that each input signal is not affected by any of the others. 2. A method of separating field wiring circuitry from logic level circuitry, typically done optically.

item

1. A category displayed on the teach pendant on a menu. 2. A set of adjacent digits, bits, or characters that is treated as a unit and conveys a single unit of information. 3. Any unique manufactured or purchased part or assembly: end product, assembly, subassembly, component, or raw material.

J**jog coordinate systems**

Coordinate systems that help you to move the robot more effectively for a specific application. These systems include JOINT, WORLD, TOOL, and USER.

JOG FRAME

A jog coordinate system you define to make the robot jog the best way possible for a specific application. This can be different from world coordinate frame.

jogging

Pressing special keys on the teach pendant to move the robot.

jog speed

Is a percentage of the maximum speed at which you can jog the robot.

joint

1. A single axis of rotation. There are up to six joints in a robot arm (P-155 swing arm has 8). 2. A jog coordinate system in which one axis is moved at a time.

JOINT

A motion type in which the robot moves the appropriate combination of axes independently to reach a point most efficiently. (Point to point, non-linear motion).

joint interpolated motion

A method of coordinating the movement of the joints so all joints arrive at the desired location at the same time. This method of servo control produces a predictable path regardless of speed and results in the fastest cycle time for a particular move. Also called joint motion.

K**K**

Abbreviation for kilo, or exactly 1024 in computer jargon. Related to 1024 words of memory.

KAREL

The programming language developed for robots by the FANUC America Corporation.

L**label**

An ordered set of characters used to symbolically identify an instruction, a program, a quantity, or a data area.

LCD

See liquid crystal display.

lead time

The span of time needed to perform an activity. In the production and inventory control context, this activity is normally the procurement of materials and/or products either from an outside supplier or from one's own manufacturing facility. Components of lead time can include order preparation time, queue time, move or transportation time, receiving and inspection time.

LED

See Light Emitting Diode.

LED display

An alphanumeric display that consists of an array of LEDs.

Light Emitting Diode

A solid-state device that lights to indicate a signal on electronic equipment.

limiting device

A device that restricts the work envelope by stopping or causing to stop all robot motion and that is independent of the control program and the application programs.

limit switch

A switch that is actuated by some part or motion of a machine or equipment to alter the electrical circuit associated with it. It can be used for position detection.

linear

A motion type in which the appropriate combination of axes move in order to move the robot TCP in a straight line while maintaining tool center point orientation.

liquid crystal display

A digital display on the teach pendant that consists of two sheets of glass separated by a sealed-in, normally transparent, liquid crystal material. Abbreviated LCD.

little-endian

The adjectives big-endian and little-endian refer to which bytes are most significant in multi-byte data types and describe the order in which a sequence of bytes is stored in a computer's memory. In a big-endian system, the most significant value in the sequence is stored at the lowest storage address (i.e., first). In a little-endian system, the least significant value in the sequence is stored first.

load

1. The weight (force) applied to the end of the robot arm. 2. A device intentionally placed in a circuit or connected to a machine or apparatus to absorb power and convert it into the desired useful form. 3. To copy programs or data into memory storage.

location

1. A storage position in memory uniquely specified by an address. 2. The coordinates of an object used in describing its x, y, and z position in a Cartesian coordinate system.

lockout/tagout

The placement of a lock and/or tag on the energy isolating device (power disconnecting device) in the off or open position. This indicates that the energy isolating device or the equipment being controlled will not be operated until the lock/tag is removed.

log

A record of values and/or action for a given function.

logic

A fixed set of responses (outputs) to various external conditions (inputs). Also referred to as the program.

loop

The repeated execution of a series of instructions for a fixed number of times, or until interrupted by the operator.

M

mA

See milliamper.

machine language

A language written in a series of bits that are understandable by, and therefore instruct, a computer. This is a "first level" computer language, as compared to a "second level" assembly language, or a "third level" compiler language.

machine lock

A test run option that allows the operator to run a program without having the robot move.

macro

A source language instruction from which many machine-language instructions can be generated.

magnetic disk

A metal or plastic floppy disk (not supported on version 7.10 and later) that looks like a phonograph record whose surface can store data in the form of magnetized spots.

magnetic disk storage

A storage device or system consisting of magnetically coated metal disks.

magnetic tape

Plastic tape, like that used in tape recorder, on which data is stored in the form of magnetized spots.

maintenance

Keeping the robots and system in their proper operating condition.

MC

See memory card.

mechanical unit

The robot arm, including auxiliary axis, and hood/deck and door openers.

medium

plural **media** . The physical substance upon which data is recorded, such as a memory card (or floppy disk which is not supported on version 7.10 and later).

memory

A device or media used to store information in a form that can be retrieved and is understood by the computer or controller hardware. Memory on the controller includes C-MOS RAM, Flash ROM and D-RAM.

memory card

A C-MOS RAM memory card or a flash disk-based PC card.

menu

A list of options displayed on the teach pendant screen.

message

A group of words, variable in length, transporting an item of information.

microprocessor

A single integrated circuit that contains the arithmetic, logic, register, control and memory elements of a computer.

microsecond

One millionth (0.000001) of a second

milliampere

One one-thousandth of an ampere. Abbreviated mA.

millisecond

One thousandth of a second. Abbreviated msec.

module

A distinct and identifiable unit of computer program for such purposes as compiling, loading, and linkage editing. It is eventually combined with other units to form a complete program.

motion type

A feature that allows you to select how you want the robot to move from one point to the next. MOTYPES include joint, linear, and circular.

mode

1. One of several alternative conditions or methods of operation of a device. 2. The most common or frequent value in a group of values.

N**network**

1. The interconnection of a number of devices by data communication facilities. "Local networking" is the communications network internal to a robot. "Global networking" is the ability to provide communications connections outside of the robot's internal system. 2. Connection of geographically separated computers and/or terminals over communications lines. The control of transmission is managed by a standard protocol.

non-volatile memory

Memory capable of retaining its stored information when power is turned off.

O

Obstacle Model.

Used in Interference Checking, the Obstacle Model is the set of virtual model elements (spheres, cylinders, and planes) that are used to represent the shape and the location of a given obstacle in space.

off-line

Equipment or devices that are not directly connected to a communications line.

off-line operations

Data processing operations that are handled outside of the regular computer program. For example, the computer might generate a report off-line while the computer was doing another job.

off-line programming

The development of programs on a computer system that is independent of the "on-board" control of the robot. The resulting programs can be copied into the robot controller memory.

offset

The count value output from a A/D converter resulting from a zero input analog voltage. Used to correct subsequent non-zero measurements also incremental position or frame adjustment value.

on-line

A term to describe equipment or devices that are connected to the communications line.

on-line processing

A data processing approach where transactions are entered into the computer directly, as they occur.

operating system

Lowest level system monitor program.

operating work envelope

The portion of the restricted work envelope that is actually used by the robot while it is performing its programmed motion. This includes the maximum the end-effector, the workpiece, and the robot itself.

operator

A person designated to start, monitor, and stop the intended productive operation of a robot or robot system.

operator box

A control panel that is separate from the robot and is designed as part of the robot system. It consists of the buttons, switches, and indicator lights needed to operate the system.

operator panel

A control panel designed as part of the robot system and consisting of the buttons, switches, and indicator lights needed to operate the system.

optional features

Additional capabilities available at a cost above the base price.

OR

An operation that places two contacts or groups of contacts in parallel. Any of the contacts can control the resultant status, also a mathematical operation.

orientation

The attitude of an object in space. Commonly described by three angles: rotation about x (w), rotation about y (p), and rotation about z (r).

origin

The point in a Cartesian coordinate system where axes intersect; the reference point that defines the location of a frame.

OT

See overtravel.

output

Information that is transferred from the CPU for control of external devices or processes.

output device

A device, such as starter motors, solenoids, that receive data from the programmable controller.

output module

An I/O module that converts logic levels within the CPU to a usable output signal for controlling a machine or process .

outputs

Signals, typically on or off, that controls external devices based upon commands from the CPU.

override

See general override.

overtravel

A condition that occurs when the motion of a robot axis exceeds its prescribed limits.

overwrite

To replace the contents of one file with the contents of another file when copying.

P**parity**

The anticipated state, odd or even, of a set of binary digits.

parity bit

A binary digit added to an array of bits to make the sum of all bits always odd or always even.

parity check

A check that tests whether the number of ones (or zeros) in an array of binary digits is odd or even.

parity error

A condition that occurs when a computed parity check does not agree with the parity bit.

part

A material item that is used as a component and is not an assembly or subassembly.

pascal

A unit of pressure in the meter-kilogram-second system equivalent to one newton per square meter.

path

1. A variable type available in the KAREL system that consists of a list of positions. Each node includes positional information and associated data. 2. The trajectory followed by the TCP in a move.

PCB

See printed circuit board.

PC Interface

The PC Interface option provides the RPC functions and PC send macros required by applications created using PC Developer's Kit.

pendant

See teach pendant.

PLC

See programmable logic controller or cell controller.

PMC

The programmable machine controller (PMC) functions provide a ladder logic programming environment to create PMC functions. This provides the capability to use the robot I/O system to run PLC programs in the background of normal robot operations. This function can be used to control bulk supply systems, fixed automation that is part of the robot workcell, or other devices that would normally require basic PLC controls.

printed circuit board

A flat board whose front contains slots for integrated circuit chips and connections for a variety of electronic components, and whose back is printed with electrically conductive pathways between the components.

production mode

See automatic mode.

program

1. A plan for the solution of a problem. A complete program includes plans for the transcription of data, coding for the computer, and plans for the absorption of the results into the system. 2. A sequence of instructions to be executed by the computer or controller to control a robot/robot system. 3. To furnish a computer with a code of instructions. 4. To teach a robot system a specific set of movements and instructions to do a task.

programmable controller

See programmable logic controller or cell controller.

programmable logic controller

A solid-state industrial control device that receives inputs from user-supplied control devices, such as switches and sensors, implements them in a precise pattern determined by ladder diagram-based programs stored in the user memory, and provides outputs for control of processes or user-supplied devices such as relays and motor starters.

Program ToolBox

The Program ToolBox software provides programming utilities such as mirror image and flip wrist editing capabilities.

protocol

A set of hardware and software interfaces in a terminal or computer that allows it to transmit over a communications network, and that collectively forms a communications language.

psi

Pounds per square inch.

Q**queue.**

1. Waiting lines resulting from temporary delays in providing service. 2. The amount of time a job waits at a work center before set-up or work is performed on the job. See also job queue.

R**RAM**

See Random Access Memory.

random access

A term that describes files that do not have to be searched sequentially to find a particular record but can be addressed directly.

Random Access Memory

1. Volatile, solid-state memory used for storage of programs and locations; battery backup is required. 2. The working memory of the controller. Programs and variable data must be loaded into RAM before the program can execute or the data can be accessed by the program.

range

1. A characterization of a variable or function. All the values that a function can possess. 2. In statistics, the spread in a series of observations. 3. A programmable voltage or current spectrum of values to which input or output analog signals can be limited.

RI

Robot input.

RO

Robot output.

read

To copy, usually from one form of storage to another, particularly from external or secondary storage to internal storage. To sense the meaning of arrangements of hardware. To sense the presence of information on a recording medium.

Read Only Memory

A digital memory containing a fixed pattern of bits that you cannot alter.

record

To store the current set or sets of information on a storage device.

recovery

The restoration of normal processing after a hardware or software malfunction through detailed procedures for file backup, file restoration, and transaction logging.

register

1. A special section of primary storage in a computer where data is held while it is being worked on.
2. A memory device capable of containing one or more computer bits or words.

remote/local

A device connection to a given computer, with remote devices being attached over communications lines and local devices attached directly to a computer channel; in a network, the computer can be a remote device to the CPU controlling the network.

repair

To restore robots and robot systems to operating condition after damage, malfunction, or wear.

repeatability

The closeness of agreement among the number of consecutive movements made by the robot arm to a specific point.

reset

To return a register or storage location to zero or to a specified initial condition.

restricted work envelope

That portion of the work envelope to which a robot is restricted by limiting devices that establish limits that will not be exceeded in the event of any reasonably foreseeable failure of the robot or its controls. The maximum distance the robot can travel after the limited device is actuated defines the restricted work envelope of the robot.

RIA

Robotic Industries Association Subcommittee of the American National Standards Institute, Inc.

robot

A reprogrammable multifunctional manipulator designed to move material, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of tasks.

Robot Model.

Used in Interference Checking, the Robot Model is the set of virtual model elements (sphere and cylinders) that are used to represent the location and shape of the robot arm with respect to the robot's base. Generally, the structure of a six axes robot can be accurately modeled as a series of cylinders and spheres. Each model element represents a link or part of the robot arm.

ROM

See Read Only Memory.

routine

1. A list of coded instructions in a program. 2. A series of computer instructions that performs a specific task and can be executed as often as needed during program execution.

S**saving data.**

Storing program data in Flash ROM, to a floppy disk (not supported on version 7.10 and later), or memory card.

scfm

Standard cubic feet per minute.

scratch start

Allows you to enable and disable the automatic recovery function.

sensor

A device that responds to physical stimuli, such as heat, light, sound pressure, magnetism, or motion, and transmits the resulting signal or data for providing a measurement, operating a control or both. Also a device that is used to measure or adjust differences in voltage in order to control sophisticated machinery dynamically.

serial communication

A method of data transfer within a PLC whereby the bits are handled sequentially rather than simultaneously as in parallel transmission.

serial interface

A method of data transmission that permits transmitting a single bit at a time through a single line. Used where high speed input is not necessary.

Server Side Include (SSI)

A method of calling or "including" code into a web page.

servomotor

An electric motor that is controlled to produce precision motion. Also called a "smart" motor.

SI

System input.

signal

The event, phenomenon, or electrical quantity that conveys information from one point to another.

significant bit

A bit that contributes to the precision of a number. These are counted starting with the bit that contributes the most value, of "most significant bit", and ending with the bit that contributes the least value, or "least significant bit".

singulating

Separating parts into a single layer.

slip sheet

A sheet of material placed between certain layers of a unit load. Also known as tier sheet.

SO

System output.

specific gravity

The ratio of a mass of solid or liquid to the mass of an equal volume of water at 45C. You must know the specific gravity of the dispensing material to perform volume signal calibration. The specific gravity of a dispensing material is listed on the MSDS for that material.

SRAM

A read/write memory in which the basic memory cell is a transistor. SRAM (or S-RAM) tends to have a lower density than DRAM. A battery can be used to retain the content upon loss of power.

slpm

Standard liters per minute.

Standard Operator Panel (SOP).

A panel that is made up of buttons, keyswitches, and connector ports.

state

The on or off condition of current to and from an input or output device.

statement

See instruction.

storage device

Any device that can accept, retain, and read back one or more times. The available storage devices are SRAM, Flash ROM (FROM or F-ROM), floppy disks (not available on version 7.10 and later), memory cards, or a USB memory stick.

system variable

An element that stores data used by the controller to indicate such things as robot specifications, application requirements, and the current status of the system.

T**Tare**

The difference between the gross weight of an object and its contents, and the object itself. The weight of an object without its contents.

TCP

See tool center point.

teaching

Generating and storing a series of positional data points effected by moving the robot arm through a path of intended motions.

teach mode

1. The mode of operation in which a robot is instructed in its motions, usually by guiding it through these motions using a teach pendant. 2. The generation and storage of positional data. Positional data can be taught using the teach pendant to move the robot through a series of positions and recording those positions for use by an application program.

teach pendant

1. A hand-held device used to instruct a robot, specifying the character and types of motions it is to undertake. Also known as teach box, teach gun. 2. A portable device, consisting of an LCD display and a keypad, that serves as a user interface to the KAREL system and attaches to the operator box or operator panel via a cable. The teach pendant is used for robot operations such as jogging the robot, teaching and recording positions, and testing and debugging programs.

telemetry

The method of transmission of measurements made by an instrument or a sensor to a remote location.

termination type

Feature that controls the blending of robot motion between segments.

tool

A term used loosely to define something mounted on the end of the robot arm, for example, a hand, gripper, or an arc welding torch.

tool center point

1. The location on the end-effector or tool of a robot hand whose position and orientation define the coordinates of the controlled object. 2. Reference point for position control, that is, the point on the tool that is used to teach positions. Abbreviated TCP.

TOOL Frame

The Cartesian coordinate system that has the position of the TCP as its origin to set. The z-axis of the tool frame indicates the approach vector for the tool.

TP.

See teach pendant.

transducer

A device for converting energy from one form to another.

U**UOP**

See user operator panel.

URL

Universal Resource Locator. A standard addressing scheme used to locate or reference files on web servers.

USB memory stick

The controller USB memory stick interface supports a USB 1.1 interface. The USB Organization specifies standards for USB 1.1 and 2.0. Most memory stick devices conform to the USB 2.0 specification for operation and electrical standards. USB 2.0 devices as defined by the USB Specification must be backward compatible with USB 1.1 devices. However, FANUC America Corporation does not support any security or encryption features on USB memory sticks. The controller supports most widely-available USB Flash memory sticks from 32MB up to 1GB in size.

USER Frame

The Cartesian coordinate system that you can define for a specific application. The default value of the User Frame is the World Frame. All positional data is recorded relative to User Frame.

User Operator Panel

User-supplied control device used in place of or in parallel with the operator panel or operator box supplied with the controller. Abbreviated UOP .

V**variable**

A quantity that can assume any of a given set of values.

variance

The difference between the expected (or planned) and the actual, also statistics definitions.

vision system

A device that collects data and forms an image that can be interpreted by a robot computer to determine the position or to “see” an object.

volatile memory

Memory that will lose the information stored in it if power is removed from the memory circuit device.

W**web server**

An application that allows you to access files on the robot using a standard web browser.

warning device

An audible or visible device used to alert personnel to potential safety hazards.

work envelope

The volume of space that encloses the maximum designed reach of the robot manipulator including the end effector, the workpiece, and the robot itself. The work envelope can be reduced or restricted by limiting devices. The maximum distance the robot can travel after the limit device is actuated is considered the basis for defining the restricted work envelope.

write

To deliver data to a medium such as storage.

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